

Robot-assisted navigation with adaptive impedance and path planning for visually occluded users

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Abstract

Considering the number of visually impaired individuals worldwide and the limited availability of assistance for safe navigation, there is a growing need for intelligent robotic systems that can provide reliable guidance. This manuscript presents an adaptive 3D human guidance approach using a mobile manipulator to assist visually impaired users as they navigate unstructured spaces. The proposed system generates and executes a collision-free path toward a designated goal while avoiding both static and dynamic obstacles at varying heights for the entire system, which comprises of an omnidirectional mobile base, a torque-controlled robotic arm, and the physically coupled user interacting with the arm's end-effector. To ensure the full-body safety of the user, we introduce a leg tracking algorithm based on 2D LiDAR sensors integrated into the mobile base, along with a human height estimation algorithm that uses an RGB-D camera. Based on the real-time tracking of the human state, the proposed approach provides responsive guidance: *Horizontal Pulling* adjusts the robot's lateral motion and base velocity according to the user's leg position when deviations from the collision-free trajectory occur, while *Vertical Pulling* modulates the arm's height to prevent upper-body collisions, based on estimated user height and the obstacle information from an attached 3D LiDAR. Additionally, an *Impedance Tuning* algorithm is designed to dynamically adjust impedance parameters in coordination with the pulling force based on the estimated collision risk level. The effectiveness of the proposed approach is demonstrated through extensive multi-subject user studies in controlled environments and a proof-of-concept deployment in a real-world scenario.

Keywords

physical human–robot interaction (pHRI), assistive robotics, visually impaired, mobile manipulation, adaptive impedance control

1. Introduction

Over the past few decades, we have witnessed a shift from isolating industrial robots behind safety cages, where any form of physical contact with humans was strictly prohibited, to a new paradigm in which robots and humans can operate side-by-side, engaging even in direct physical interaction (Hägele et al., 2016). This evolution has been largely driven by a growing demand for robotic systems that can assist humans in unstructured, dynamic environments where preprogramming and isolation are no longer feasible (Ajoudani et al., 2018b).

Today, physical human–robot interaction (pHRI) plays a pivotal role across a wide range of applications. A notable example is rehabilitation robotics, where direct contact between humans and robots exists (Qassim and Wan Hasan, 2020; Zhang et al., 2020; Ruiz-Ruiz et al., 2021). In flexible manufacturing environments, pHRI enables collaborative

transportation of payloads (Sirintuna et al., 2023a, 2023b, 2024b; Sirintuna et al., 2023b; Sirintuna et al., 2024) and shared heavy-duty operations (Giammarino et al., 2024). In addition to these, the domain of search-and-rescue benefits from physical interaction for tasks like casualty extraction (Choi et al., 2019) and underactuated limb manipulation (Gandarias et al., 2019). Another key domain

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within pHRI is assistive and healthcare robotics, recognized as one of the remarkable and rapidly growing fields (Gongor and Tutsoy, 2025), with recent developments including Ding et al. (2014), Fitter et al. (2020), Liu et al. (2018), Nakamura and Tsukagoshi (2018), Ruiz-Ruiz et al. (2022), and Su et al. (2021).

A particularly compelling application of assistive robotics is to guide visually impaired individuals through unstructured environments (Khan et al., 2021). Several studies have focused on robotized white canes as a solution to this challenge. A pioneering example of this concept was presented in Ulrich and Borenstein (2001), where the cane actively rotates toward a suitable direction upon obstacle detection by the sonar sensors. Similarly, in Shim and Yoon (2003), a robotic cane named RoJi was introduced, which incorporates real-time user input on the intended walking direction while avoiding obstacles. More recently, Ye et al. (2016) proposed leveraging a 3D camera in combination with an indoor navigation algorithm to provide guidance for blind users through enhanced environmental perception. Building upon these advancements, Slade et al. (2021) developed the Augmented Cane, which integrates a suite of sensors—including a 2D LiDAR, camera, GPS, and IMU—to significantly extend navigational assistance beyond indoor settings, enabling robust guidance in outdoor environments as well. Their study demonstrated that the proposed steering assistance not only improved user confidence but also increased walking speed.

Despite these advances, cane-based solutions remain inherently limited in versatility and autonomy. Recent work explores mobile platforms, such as wheeled or dog-like robots, which offer dynamic mobility while maintaining physical guidance (Hong et al., 2022). For example, Xiao et al. (2021) presented a quadrupedal robot with a leash to guide the visually impaired in narrow spaces while avoiding collisions. Building on this paradigm, Chen et al. (2023) introduced a comfort-driven framework that integrates a force-controlled elastic leash and a human motion model to minimize abrupt force changes during guidance. Alternatively, in Zhang et al. (2023), a wheeled robot with an aluminum handle frame was developed to guide the user while not disturbing the other people in the vicinity. Yet, to the best of our knowledge, no prior work has explored the potential of mobile robots equipped with a manipulator for navigation guidance of visually impaired users. Leveraging the loco-manipulation capabilities of such platforms could enable more effective guidance, particularly in labyrinthine indoor spaces or corridors with recesses and sharp turns, where rope-like interfaces are highly inefficient.

Beyond navigation support for visually occluded users, mobile manipulators have been employed in a variety of assistive robot applications where physical interaction is central to the task. Despite a rapid expansion in the literature on physically assistive robots (PARs) over the past decade, much of the work remains isolated within specific

application domains (Nanavati et al., 2024). Among activities of daily living (ADLs), eating is one of the crucial tasks where mobile manipulators have recently been explored for use in assisted feeding. In Park et al. (2020), a proof-of-concept feeding assistance system was developed using a mobile manipulator, incorporating compliant arm motions and anomaly detection capabilities. Alternatively, in Erickson et al. (2018), a deep learning-based MPC approach for assisted dressing was proposed, where the experiments with healthy subjects demonstrated its effectiveness while reducing the applied forces to users. For other contact-rich tasks, namely shaving and wiping, Hawkins et al. collected data on contact forces from healthy participants during these tasks (Hawkins et al., 2012), and then used this information to develop a system capable of assisting users with severe motor impairments (Hawkins et al., 2014). Alongside these functional tasks, socially assistive robots (SARs) also leverage mobile manipulators to provide physical guidance. For instance, dance-based exercises, where a compliant robot is employed as a partner, have been explored to improve the balance and gait of the elderly users (Chen et al., 2015, 2017).

Although the diverse nature of these tasks makes the simple translation of assistive strategies challenging across them, the studies emphasize that careful attention must be given to regulating physical interaction to realize the advantages of mobile manipulators. This physical link between the robot and the user must remain not only safe, but also responsive to dynamic human behavior (Duchaine and Gosselin, 2009). One common method used in the literature is impedance control, which allows the robot to move in a compliant and cooperative way by adjusting how it reacts to forces at the point of contact (Hogan, 1984). However, fixed impedance parameters often fall short in real-world assistive scenarios, where user intent and human movement can vary significantly (Ferraguti et al., 2013). To address this issue, recent research has focused on adaptive impedance control, which dynamically adjusts the system's behavior based on real-time feedback about the environment and human state (Ajoudani et al., 2014; Balatti et al., 2020). Indeed, robots with lower stiffness are frequently perceived as safer by humans, since they give the impression of being more adaptable to unexpected situations without posing a risk (Rubagotti et al., 2022). Furthermore, lower stiffness can promote a greater sense of control among individuals interacting with the robot (Fitter and Kuchenbecker, 2020). However, in situations that require precise tracking, increasing the stiffness of the robot enhances accuracy (Abu-Dakka and Saveriano, 2020), though it may reduce the subjective feeling of autonomy by the user.

In this article, we propose an adaptive 3D pHRI system that enables a mobile manipulator to guide a visually impaired user through an unstructured environment by planning and executing safe trajectories while actively avoiding obstacles at varying heights (see Figure 1). One key

advantage of using mobile robots in assistive scenarios is their ability to support a broad range of daily living tasks thanks to their combined locomotion and manipulation capabilities. For example, the mobile base can assist with transporting heavy items, while the arm can be used to open doors or retrieve objects that are otherwise out of reach. Our proposed approach is designed to promote greater autonomy for visually impaired individuals by offering safe navigation, and this platform is meant to support the users in carrying out everyday routines. In particular, this study presents the following contributions:

- Development of a human guidance planner that ensures collision-free trajectories for a coupled human–robot system consisting of a mobile base, robotic arm, and a physically linked simulated impaired user.
- Development of an adaptive 3D pulling motion assistance that provides physical guidance to help users avoid collisions with both static and dynamic obstacles. The *Horizontal Pulling* unit adjusts both the lateral position of the arm end-effector and the velocity of the mobile base based on human leg tracking via 2D LiDARs, when the users deviate from the intended safe trajectory. In parallel, the *Vertical Pulling* unit prevents upper-body collisions by modulating the height of the end-effector using human height estimated from an RGB-D camera and obstacle information from an attached 3D LiDAR.
- Design of an *Impedance Tuning* algorithm that independently regulates impedance parameters along all three spatial axes in real time, based on tracked human motion and detected environmental obstacles. When the visually impaired user deviates from the planned trajectory and a potential collision is identified, the algorithm works in coordination with the pulling units to adaptively increase the arm’s stiffness in the corresponding direction, proportionally to the estimated level of risk.
- Experimentation and validation of the overall system through both user studies in controlled laboratory settings and a proof-of-concept demonstration in a real-world environment. The user studies specifically evaluated the system’s performance under varying obstacle heights. Both quantitative metrics and qualitative evaluations through a usability questionnaire are presented to assess the effectiveness and user experience of the system.

The preliminary version of this study was introduced in our previous conference paper (Balatti et al., 2024). However, significant contributions have been added to this article with respect to the original work. Most notably, we introduced a novel *Vertical Pulling* unit that elevates the concept of guidance to full-body safety by addressing the

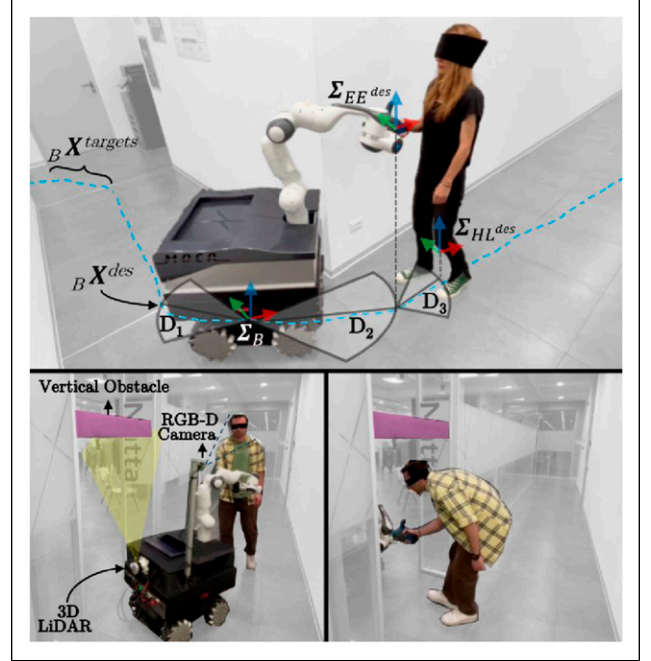


Figure 1. We propose an adaptive 3D human guidance approach using a mobile manipulator to ensure the full-body safety of the visually impaired users while navigating around both static and dynamic obstacles in complex environments.

often-overlooked risk of upper-body collisions. While the earlier approach was vulnerable to upper-level obstacles, as it relied solely on ground-level sensing with 2D LiDARs positioned at the base, this new unit enables spatially-aware guidance across all three dimensions, significantly enhancing the robot’s ability to protect the user from obstacles at any height. To the best of our knowledge, this work is the first to explore vertical physical guidance for avoiding upper-body collisions. Additionally, this article introduces a method to estimate human height in real time using an RGB-D camera, which is essential for enabling safe and effective *Vertical Pulling*. Finally, the enhanced system is validated through additional experiments not included in the previous version. These new scenarios incorporate obstacles at varying heights, as well as dynamic obstacles (e.g., walking pedestrians), posing realistic challenges to blindfolded participants. The effectiveness of the presented approach is evaluated through the quantitative and qualitative results of the user studies.

Notably, although this paper focuses on robot-assisted navigation for the (simulated) visually impaired, the proposed approach has broader applicability. For example, in post-disaster scenarios where individuals are trapped in cluttered and risky environments, such systems, even those equipped with legged locomotion for uneven terrains, could be deployed to safely guide or transport them to secure areas. Beyond rescue operations, similar systems could also

support elderly care, temporary mobility loss, or navigation assistance in complex indoor environments such as hospitals or airports.

The remainder of this article is organized as follows: Section 2 introduces the proposed adaptive 3D human guidance approach, structured around three core modules: *Human Tracking*, *Planning*, and *Control*. Section 3 details the hardware setup and the software architecture. In Section 4, we describe the conducted user studies under varying obstacle conditions and an additional proof-of-concept demonstration in a real-world scenario. Finally, Section 5 discusses the overall performance of the proposed system and concludes the paper.

2. 3D robot-assisted guidance approach

The primary objective of the proposed approach is to assist individuals in navigating complex environments using a mobile manipulator. The collaborative robot allows users to grasp the underactuated robotic hand at the end-effector, providing guidance toward a designated goal while avoiding both static and dynamic obstacles at varying heights. To achieve this, a 3D adaptive pulling strategy has been implemented that continuously adjusts both the end-effector's position and the parameters of the Cartesian impedance controller. This ensures real-time correction of the human's trajectory, keeping them within a safe, collision-free path generated by the system.

The overall architecture of the presented approach (see Figure 2) can be categorized into three main modules: (1) the *Human Tracking* module that monitors both the head and legs of the human with respect to the robot, (2) the *Planning* module that produces adaptive pulling actions, when necessary, toward collision-free path which is generated based on the detected obstacles including the ones that can hit the upper body of the human, and (3) the *Control* module that generates desired joint torques and velocities for the mobile manipulator.

2.1. Human tracking

Real-time tracking of the human is crucial for safe guidance to avoid both ground-level obstacles and those that may hit the upper body of the human, as illustrated in Figure 2. Although mobile manipulators can be equipped with numerous sensory systems, monitoring the whole human body may not be feasible due to factors such as sensor limitations and occlusions. To address this challenge, we selected to utilize the 2D LiDAR sensor in the mobile base for the legs of the human, while an RGB-D camera is employed to obtain the human height on the fly.

2.1.1. Legs tracker. Laser scan-based approaches stand out as suitable solutions for human leg tracking due to their wide field of view and real-time performance. To eliminate both

the noise and background objects present in these approaches, our implementation begins by filtering the 2D laser scan using a pass-through filter based on angle and distance ranges, as described below:

$$\mathcal{P}^{filt} = \{s_i = (s_{i,\rho}, s_{i,\phi}) \mid \rho^{min} \leq s_{i,\rho} \leq \rho^{max}, \phi^{min} \leq s_{i,\phi} \leq \phi^{max}\}, \quad (1)$$

where $(s_{i,\rho}, s_{i,\phi})$ are the polar coordinates of the i -th point in the 2D laser scan (s_i), with $s_{i,\rho} \in \mathbb{R}_{\geq 0}$ being the radial distance and $s_{i,\phi} \in \mathbb{S}^1$ the angle of the point relative to the sensor. $\phi^{min} \in \mathbb{S}$ and $\phi^{max} \in \mathbb{S}$ are the angular limits of the filter, while $\rho^{min} \in \mathbb{R}_{\geq 0}$ and $\rho^{max} \in \mathbb{R}_{> 0}$ are the radial limits. In particular, we set the radial limits to $\rho^{min} = 0$ m and $\rho^{max} = 1.5$ m, and the angular limits to $\phi^{min} = -1.9$ and $\phi^{max} = 0.6$ radians. These limits were selected to cover the region where the human is typically located while holding the robot's hand during guidance. Once the data is filtered, we convert the polar coordinates of the filtered data ($\mathcal{P}^{filt}$) to Cartesian coordinates for further processing using the following equations:

$$s_{i,x} = s_{i,\rho} \cos(s_{i,\phi}), \quad (2)$$

$$s_{i,y} = s_{i,\rho} \sin(s_{i,\phi}), \quad (3)$$

where $s_{i,x} \in \mathbb{R}$ and $s_{i,y} \in \mathbb{R}$ are the Cartesian coordinates of $s_i \in \mathbb{R}^2$ in $\mathcal{P}^{filt} \subset \mathbb{R}^2$.

Following the conversion to Cartesian coordinates, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm detects clusters in the 2D point cloud, identifying potential objects. Each cluster centroid is computed as the mean position of its points:

$$\mathbf{m}_k = \left(\frac{1}{|\mathcal{C}_k|} \sum_{s_j \in \mathcal{C}_k} s_{j,x}, \frac{1}{|\mathcal{C}_k|} \sum_{s_j \in \mathcal{C}_k} s_{j,y} \right), \quad (4)$$

where $\mathcal{C}_k \subset \mathbb{R}^2$ is the set of points in the k -th cluster, $|\mathcal{C}_k| \in \mathbb{N}_{> 0}$ is the number of points in the cluster, and $\mathbf{m}_k = (m_{k,x}, m_{k,y}) \in \mathbb{R}^2$ is the coordinates of the centroid.

In order to maintain consistency in tracking over time while both the human and robot move, a centroid tracking algorithm is employed, which associates the found clusters for each leg with the tracked ones. In this algorithm, a coherence threshold ($d^{th} \in \mathbb{R}_{> 0}$) is used, which defines the maximum allowed distance between a new centroid $\mathbf{m}_k(t)$ at time t and an existing centroid $\mathbf{m}_k(t-1)$ from the previous time step $t-1$. The distance between the centroids is computed as:

$$\|\mathbf{m}_k(t) - \mathbf{m}_k(t-1)\| = \sqrt{(m_{k,x}(t) - m_{k,x}(t-1))^2 + (m_{k,y}(t) - m_{k,y}(t-1))^2}. \quad (5)$$

If this distance between centroids is smaller than the coherence threshold ($\|\mathbf{m}_k(t) - \mathbf{m}_k(t-1)\| < d^{th}$), the

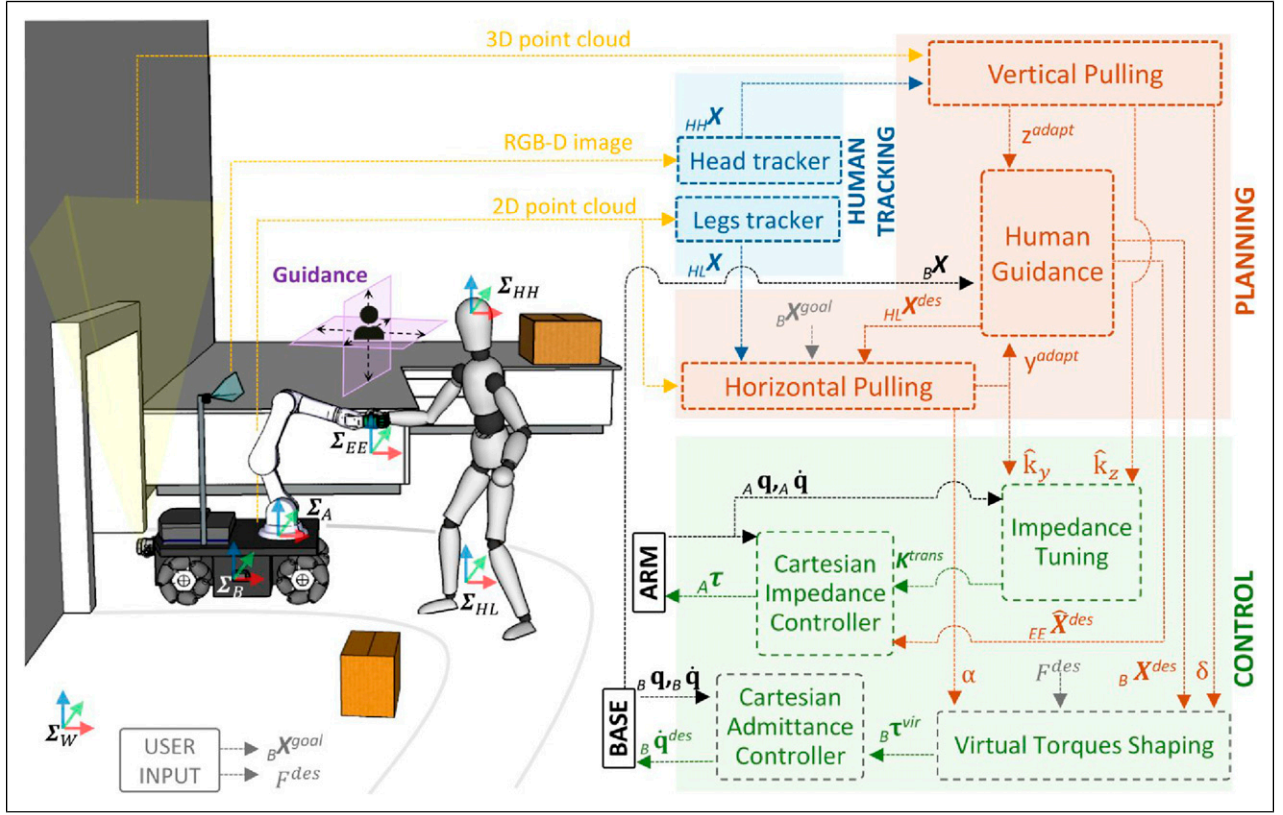


Figure 2. A high-level overview of the proposed system. The system can be structured into three main modules: (1) *Human Tracking* (blue), which monitors both the user’s leg and head positions relative to the robot using onboard sensors; (2) *Planning* (orange), which computes adaptive pulling actions toward a collision-free path that is generated by taking into account both lower- and upper-body obstacles; and (3) *Control* (green), which generates the corresponding joint torques and base velocities for the mobile manipulator. Each unit is implemented as a ROS node, and communication between them is handled via ROS topics, as indicated by the dotted lines.

centroids are considered the same object. The centroids that do not meet this criterion are either discarded or replaced with the previous state to maintain stability. For our application, where no significant high accelerations are expected in human motion, we chose to set the d^{th} to 0.2 m. This threshold ensures that tracking remains stable, preventing misidentification due to noise while allowing for natural human motion variations. Finally, the current human leg pose frame (Σ_{HL}) is computed as the average of the two closest centroids.

2.1.2. Head tracker. Continuous human height tracking is essential for keeping the user within the collision-free zone and preventing potential upper-body contact through vertical guidance. To achieve this, we use an RGB-D camera in combination with YOLO (You Only Look Once) (Jocher and Qiu, 2024) for real-time face detection and extraction of its bounding box. Since multiple people may be present in the camera’s field of view while the robot guides the user, the depth information provided by the camera is first used to identify the closest detected face before proceeding with height estimation. Once identified, the closest detected face that corresponds to the user is mapped to a 3D point in the camera frame (${}^C_F \mathbf{r} \in \mathbb{R}^3$) using the intrinsic camera

calibration parameters and the associated depth information of the center of the bounding box ($z^{\text{center}} \in \mathbb{R}_{>0}$), as described below:

$${}^C_F r_x = z^{\text{center}}(u - c_x)/f_{l_x}, \quad (6)$$

$${}^C_F r_y = z^{\text{center}}(v - c_y)/f_{l_y}, \quad (7)$$

$${}^C_F r_z = z^{\text{center}}, \quad (8)$$

where $(u, v) \in \mathbb{R}^2$ are the pixel coordinates of the center of the bounding box in the image plane, $(f_{l_x}, f_{l_y}) \in \mathbb{R}_{>0}^2$ are the focal lengths of the camera in the x and y directions, respectively, $(c_x, c_y) \in \mathbb{R}^2$ are the principal point coordinates of the camera.

This 3D point in the camera frame (${}^C_F \mathbf{r}$) is then transformed into the robot’s base frame (Σ_B) using the rotation matrix (${}^B_C \mathbf{R} \in \mathbb{R}^{3 \times 3}$) and the translation vector (${}^B_C \mathbf{t} \in \mathbb{R}^3$), which represent the camera frame (Σ_C) relative to the base frame (Σ_B):

$${}^B_F \mathbf{r} = {}^B_C \mathbf{R} {}^C_F \mathbf{r} + {}^B_C \mathbf{t}. \quad (9)$$

By transforming the detected face position into the robot’s base frame, the user’s height can be continuously

tracked relative to the ground. To estimate the current human height, represented by a coordinate frame Σ_{HH} located on top of the head, a fixed offset ($\lambda \in \mathbb{R}_{>0}$) is added to the z-component of the face position (${}^B_F\mathbf{r}$), as follows:

$${}^B_{HH}\mathbf{r} = {}^B_F\mathbf{r} + [0 \ 0 \ \lambda], \quad (10)$$

where the λ is selected as 15 cm to account for the average distance between the face and the top of the head. Finally, to ensure smooth and stable height tracking while mitigating abrupt fluctuations caused by potential YOLO mis-detections, the recent height measurements are averaged over a sliding time window of 0.5 s prior to being used for vertical guidance.

2.2. Planning

This module ensures that the human-robot dyad follows a collision-free and efficient path toward the desired target while remaining capable of adapting to changes in complex environments. The *Human Guidance* unit generates this safe path in real time by combining the global route toward the goal with a local plan derived from obstacles detected using the 2D LiDAR. Meanwhile, the *Horizontal Pulling* and *Vertical Pulling* units provide adaptive pulling actions to proactively guide the user back to the collision-free trajectory.

2.2.1. Human guidance. The purpose of this unit is to compute obstacle-free routes for the mobile base, the end-effector of the arm, and the human counterpart, as outlined in Algorithm 1. To achieve this, first, collision-free waypoints for the base (${}_B\mathbf{X}^{targets} \subset \mathbb{R}^{4 \times 4}$) are generated using the Robot Operating System's *move_base* (Marder-Eppstein, 2010) package. It employs a Global Planner together with the timed elastic band (TEB) local planner (Roesmann et al., 2012) to reach the base goal pose (${}_B\mathbf{X}^{goal} \in \mathbb{R}^{4 \times 4}$). In order to ensure that the mobile base avoids both static and dynamic obstacles, the algorithm replans the waypoint sequence (${}_B\mathbf{X}^{targets}$) using a local cost map, which is a grid-based representation of the environment in which each cell is assigned a traversal cost. This cost map is continuously updated from odometry (${}_B\mathbf{X} \in \mathbb{R}^{4 \times 4}$) and 2D LiDAR scan data ($\mathcal{L} \subset \mathbb{R}^3$), enabling real-time adaptation to obstacles.

Following the waypoint sequence generation, the desired collision-free trajectory poses for the base (${}_B\mathbf{X}^{des} \in \mathbb{R}^{4 \times 4}$), the end-effector (${}_{EE}\mathbf{X}^{des} \in \mathbb{R}^{4 \times 4}$), and the human legs (${}_{HL}\mathbf{X}^{des} \in \mathbb{R}^{4 \times 4}$) are obtained based on ${}_B\mathbf{X}^{targets}$. Then, ${}_{EE}\mathbf{X}^{des}$ is further adjusted to ${}_{EE}\hat{\mathbf{X}}^{des} \in \mathbb{R}^{4 \times 4}$ according on the inputs from the *Horizontal* and *Vertical Planning* units. Note that this adjustment considers the human's lateral deviations from the desired path and obstacles above that could pose a collision risk along the trajectory. Algorithm 1 provides the pseudocode for the *Human Guidance* unit, outlining the calculation of ${}_B\mathbf{X}^{des}$, ${}_{EE}\hat{\mathbf{X}}^{des}$, and ${}_{HL}\mathbf{X}^{des}$, which serve as the inputs to the *Control* module.

In order to determine the next target waypoint for the base (${}_B\mathbf{X}^{des}$), we first sort the set of candidate waypoints (${}_B\mathbf{X}^{targets}$) by their Euclidean distance from the current base pose (${}_B\mathbf{X}$). This results in an ordered list (${}_B\mathbf{X}^{targets'}$) where waypoints closer to the current pose appear before those farther away. Then, ${}_B\mathbf{X}^{des}$ is selected from the sorted waypoints, ${}_B\mathbf{X}^{targets'}$, such that it is the closest point that lies at least a distance $D_1 \in \mathbb{R}_{>0}$ away from the ${}_B\mathbf{X}$ (see lines 3–8). The *Virtual Torques Shaping* unit uses this pose to compute the base's virtual torques, which are subsequently converted into commanded velocities by the *Cartesian Admittance Controller*. Unlike the conventional use of the *move_base* package, which directly generates base velocities for execution, we opt to use ${}_B\mathbf{X}^{des}$ to have greater control over the base motion. This approach enables us to make adaptive adjustments as needed, enhancing responsiveness and compliance with both the environment and the human partner.

To guarantee that the end-effector's desired pose follows the obstacle-free path, ${}_{EE}\mathbf{X}^{des}$ is chosen from the set of prior *odom* poses (${}_B\mathbf{X}_{[0, \dots, t-1]}$) by picking the one that is positioned $D_2 \in \mathbb{R}_{>0}$ away from the current base location. The z-coordinate of the end-effector is adjusted to a predefined comfortable grasping height $h^o \in \mathbb{R}_{>0}$ (see lines 9–15). The resulting ${}_{EE}\mathbf{X}^{des}$ defines the reference desired trajectory for human guidance, which does not incorporate any human tracking information. For providing adaptive human-aware assistance, the position of the robotic hand is further modified using outputs from the *Horizontal Pulling* unit (y^{adapt} , Section 2.2.2) and the *Vertical Pulling* unit (z^{adapt} , Section 2.2.3). By dynamically adapting the ${}_{EE}\mathbf{X}^{des}$ to ${}_{EE}\hat{\mathbf{X}}^{des}$ (see line 16), our method achieves to guarantee collision-free human guidance in 3D space by correcting human deviations from the safe optimal path.

The desired human leg pose (${}_{HL}\mathbf{X}^{des}$) is determined using the same selection methodology applied to the robot's end-effector and the base. Specifically, it is chosen from the previous end-effector waypoints, ${}_{EE}\mathbf{X}^{des}_{[0, \dots, t-1]}$, that is located at a suitable interaction distance $D_3 \in \mathbb{R}_{>0}$ (see Figure 1) with the user. For the z-component of ${}_{HL}\mathbf{X}^{des}$, the height of the 2D LiDAR sensors mounted on the mobile base is used, as their point cloud data serves for planar human leg tracking (see lines 17–23).

2.2.2. Horizontal pulling. This unit is designed to ensure human safety by adaptively pulling the human in the horizontal plane toward the robot's collision-free trajectory generated by the *Human Guidance* unit. To do so, a horizontal pulling vector ($\vec{\mathbf{p}} \in \mathbb{R}^2$) is generated on the fly based on the relative position between the desired human legs' frame (Σ_{HL}^{des}) and their current frame (Σ_{HL}), as follows:

$$\vec{\mathbf{p}} = [-l_x, -l_y]^T, \quad (11)$$

Algorithm 1 Human guidance**Input:** ${}_B\mathbf{X}^{goal}, {}_B\mathbf{X}, \mathcal{L}, \mathbf{y}^{adapt}, \mathbf{z}^{adapt}$ **Output:** ${}_B\mathbf{X}^{des}, {}_{EE}\hat{\mathbf{X}}^{des}, {}_{HL}\mathbf{X}^{des}$

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1:  ${}_B\mathbf{X}^{targets} = \text{GenerateWaypoints}({}_B\mathbf{X}, {}_B\mathbf{X}^{goal}, \mathcal{L})$ 
2:  ${}_B\mathbf{X}^{targets'} = \text{sort}({}_B\mathbf{X}^{targets})$ 
3: for  $target$  in  ${}_B\mathbf{X}^{targets'}$  do
4:   if  $\|target - {}_B\mathbf{X}\| \geq D_1$  then
5:      ${}_B\mathbf{X}^{des} = target$  ▷ Desired base target
6:   break for waypoint selection
7:   end if
8: end for
9: for  $odom$  in  ${}_B\mathbf{X}_{[0, \dots, t-1]}$  do
10:  if  $\|odom - {}_B\mathbf{X}\| = D_2$  then
11:     ${}_{EE}\mathbf{X}^{des} = odom$  ▷ Desired EE pose
12:     ${}_{EE}\mathbf{X}_z^{des} = h^\circ$  follows odom
13:    break for
14:  end if
15: end for
16:
17: for  $EE$  in  ${}_{EE}\mathbf{X}_{[0, \dots, t-1]}^{des}$  do
18:  if  $\|EE - {}_{EE}\mathbf{X}^{des}\| = D_3$  then
19:     ${}_{HL}\mathbf{X}^{des} = EE$  ▷ Human desired pose
20:     ${}_{HL}\mathbf{X}_z^{des} = 2D$  LiDAR height follows desired EE
21:    break for
22:  end if
23: end for

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$${}_{EE}\hat{\mathbf{X}}^{des} = \begin{bmatrix} {}_{EE}\mathbf{R}^{des} & \begin{bmatrix} {}_{EE}X_x^{des} \\ {}_{EE}X_y^{des} + y^{adapt} \\ z^{adapt} \end{bmatrix} \\ \mathbf{0}^\top & 1 \end{bmatrix} \quad \text{▷ Adaptive pulling}$$

where l_x and $l_y \in \mathbb{R}$ represent the x and y components of the translation between the desired human leg frame and the actual one, respectively. Using this vector, which represents the real-time deviation of the human from the obstacle-free trajectory in the horizontal plane, this strategy calculates both the lateral adjustment of the end-effector ($y^{adapt} \in \mathbb{R}$) and the robot torque gain ($\alpha \in \mathbb{R}_{\geq 0}$) that adjusts the mobile base velocity when the human lags behind (see Section 2.3.4 for the details).

In order to prevent the robot from exceeding its operational limits while maintaining safety during the presented human guidance strategy, the pulling vector is saturated based on the following rule:

$$\vec{p}^{sat} = \begin{cases} \vec{p}, & \text{if } \|\vec{p}\| \leq 1 \\ \frac{\vec{p}}{\|\vec{p}\|}, & \text{else} \end{cases}. \quad (12)$$

As previously explained in Section 2.2.1, while the X-axes of both Σ_{EE} and Σ_{HL}^{des} remain aligned with the robot's direction of motion, their Y-axes are perpendicular to this direction and parallel to the ground plane, illustrated in Figure 2. Consequently, the intended lateral adjustment of the end-effector pose (y^{adapt}) is directly set to:

$$y^{adapt} = \vec{p}_y^{sat} \in \mathbb{R}. \quad (13)$$

Moreover, this module adjusts the lateral stiffness ($\hat{k}_y \in \mathbb{R}_{\geq 0}$) based on a logistic function, which provides a lighter pull when the human's deviation is small and a stronger pull as the deviation increases (see Figure 3). This function varies based on the magnitude of y^{adapt} , according to the following formulation:

$$\hat{k}_y = \zeta + \frac{(\eta - \zeta)}{1 + e^{(\gamma - |y^{adapt}|)\kappa}}, \quad (14)$$

where η, ζ, γ , and $\kappa \in \mathbb{R}$ define the function's maximum and minimum asymptotes, the inflection point, and the steepness of the curve, respectively.

For the calculation of the raw torque gain of the base, the following function is used:

$$\alpha^{raw} = -\frac{\vec{p}_x^{sat}}{d^{stop}} + 1 \quad (15)$$

where $d^{stop} \in \mathbb{R}_{<0}$ is the saturation distance at which the robot's base should stop moving when the human lags behind. Next, the raw torque gain is constrained to ensure it stays within a valid range, as defined by the following piecewise function:

$$\alpha = \begin{cases} 1, & \text{if } 1 \leq \alpha^{raw} \\ \alpha^{raw}, & \text{if } 0 \leq \alpha^{raw} \leq 1 \\ 0, & \text{else} \end{cases}. \quad (16)$$

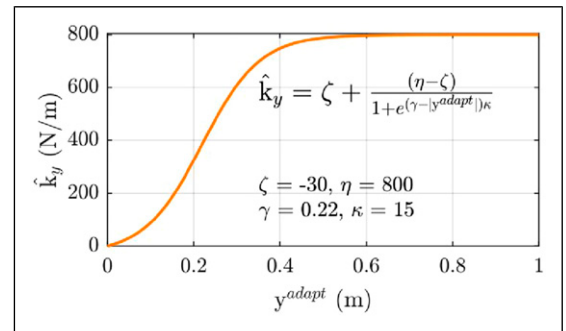


Figure 3. Lateral stiffness profile adopted by the Horizontal Pulling unit.

2.2.3. Vertical pulling. To enhance human safety in 3D space, this unit provides vertical motion assistance that helps prevent upper-body contact throughout the guidance process. This is achieved by adaptively controlling the end-effector's height (${}_{EE}\hat{X}_z^{des} = z^{adapt} \in \mathbb{R}_{>0}$), based on the tracked position of the human head and the obstacles at varying heights detected by the 3D LiDAR sensor mounted on the back of the robot (see Figure 1).

In order to obtain the obstacle information that risks potential collisions, the point cloud data (${}^L\mathcal{O} \subset \mathbb{R}^3$) in 3D LiDAR frame (Σ_L) is continuously transformed into the robot's base frame (Σ_B) during guidance, as follows:

$${}^B\mathcal{O} = {}^B_L R {}^L\mathcal{O} + {}^B_L t, \quad (17)$$

with ${}^B_L R \in \mathbb{R}^{3 \times 3}$ representing the rotation and ${}^B_L t \in \mathbb{R}^3$ denoting the translation from Σ_L to Σ_B . The resulting ${}^B\mathcal{O}$ is then spatially filtered to isolate the critical volume above the robot base, an area where any detected obstacle could pose a risk of collision with the human. This focus aligns with the objective of the *Horizontal Pulling* unit, which actively guides the human to remain within the safe corridor of the mobile base. The filtered point cloud denoted by ${}^B\mathcal{O}^{filt} \subseteq {}^B\mathcal{O}$ is defined as:

$${}^B\mathcal{O}^{filt} = \{o_i \in {}^B\mathcal{O} \mid o_{i,x} \in (\epsilon_x^{min}, \epsilon_x^{max}) \wedge o_{i,y} \in (\epsilon_y^{min}, \epsilon_y^{max}) \wedge o_{i,z} \in (\epsilon_z^{min}, \epsilon_z^{max})\}, \quad (18)$$

where each point $o_i \in \mathbb{R}^3$ is evaluated against coordinate-wise scalar boundaries $\epsilon_*^{min}, \epsilon_*^{max} \in \mathbb{R}$ (for $* \in \{x, y, z\}$) relative to Σ_B . The aforementioned region of interest (ROI) includes: (i) longitudinal bounds, with ϵ_x^{min} beginning at the base's rear edge and ϵ_x^{max} extending further back to preemptively detect potential hazards; (ii) lateral limits, where ϵ_y^{min} and ϵ_y^{max} correspond to the base's leftmost and rightmost edges; and (iii) vertical thresholds, ϵ_z^{min} and ϵ_z^{max} , covering the human interaction height range to ensure full safety. The closest obstacle, in terms of height, relative to the robot base within the filtered region, is calculated by:

$$z^{B,obs}(t) = \min(o_{i,z}^{filt}), \text{ where } o_i^{filt} \in \mathcal{O}^{filt}. \quad (19)$$

The current height measurement $z^{B,obs}(t) \in \mathbb{R}_{>0}$ is appended to the obstacle history vector of the base $\mathbf{Z}^{B,obs}$ to maintain a spatiotemporal record for vertical motion planning. To prevent outdated data from affecting decisions, entries in $\mathbf{Z}^{B,obs}$ that correspond to the previously passed positions more than $\zeta^{keep} \in \mathbb{R}_{>0}$ distance behind are discarded. Specifically, a value $z^{B,obs}(t_i)$ is removed when the cumulative distance traveled by the robot since the time t_i exceeds ζ^{keep} , as follows:

$$\int_{t_i}^t \left\| \frac{d_B \mathbf{X}(t)}{dt} \right\| dt > \zeta^{keep}. \quad (20)$$

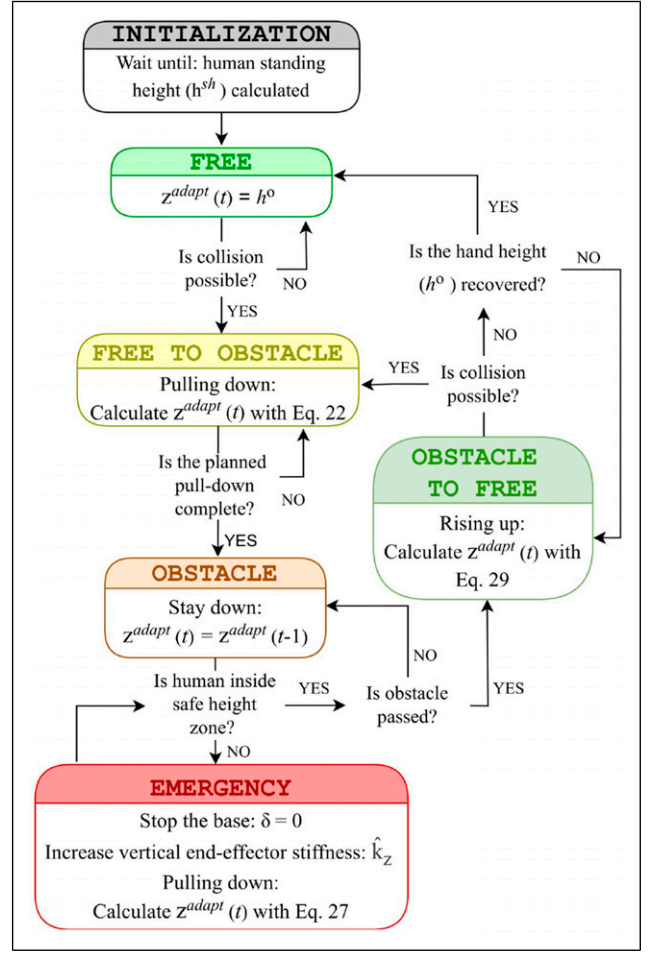


Figure 4. The state transition diagram of the *Vertical Pulling* unit. The system adapts the vertical position of the robotic arm based on real-time obstacle detection and human head tracking. Furthermore, if the collision risk increases, the system activates protective protocols that regulate both arm stiffness in the vertical axis and base motion to mitigate risks.

As a result, $\mathbf{Z}^{B,obs}$ consistently stores obstacle data within the ζ^{keep} distance. Notably, ζ^{keep} is deliberately chosen to exceed the gap between the robot base and the human when aligned, ensuring the human remains encapsulated throughout the motion.

The proposed *Vertical Pulling* unit architecture comprises five main states, namely INITIALIZATION, FREE, FREE TO OBSTACLE, OBSTACLE, and OBSTACLE TO FREE, along with an EMERGENCY state for safety-critical scenarios (see Figure 4). Before the system begins its operation, the INITIALIZATION state computes the human standing height ($h^{sh} \in \mathbb{R}_{>0}$) by averaging ${}_{HH}X_z$ input obtained from the *Head Tracker* unit over a short interval. This human standing height calculation provides a reference for safe obstacle avoidance decisions for the subsequent states. The robot begins moving only after h^{sh} is determined, maintaining a comfortable grasping end-effector height,

$$z^{adapt}(t) = h^o, \quad (21)$$

in the FREE state unless a collision risk is detected.

As the robot executes its trajectory, if an obstacle that is lower than human standing height with a safety margin ($z^{B,obs}(t) < h^{sh} + \Delta_{safe} \in \mathbb{R}_{>0}$) is detected, the system transitions to the FREE TO OBSTACLE state. Here, the *Vertical Pulling* unit initiates a controlled lowering motion of the end-effector in the following rule:

$$z^{adapt}(t) = \begin{cases} h^o - \zeta(t)\Delta_{down}/\zeta^{req} & \text{if } \zeta(t) \leq \zeta^{req} \\ h^o - \Delta_{down} & \text{else,} \end{cases} \quad (22)$$

with

$$\zeta(t) = \int_{t_{det}}^t \left\| \frac{d_B \mathbf{X}(t)}{dt} \right\| dt, \quad (23)$$

where $\zeta(t)$ corresponds to the traveled distance by the robot base since obstacle detection (t_{det}), $\Delta_{down} \in \mathbb{R}_{>0}$ is the selected downward pulling length, and $\zeta^{req} \in \mathbb{R}_{>0}$ is the required distance for the mobile base while executing the planned downward motion. Note that, ζ^{req} should be selected smaller than the distance between the $_B \mathbf{X}$ and $_{EE} \mathbf{X}^{des}$ frames in the X - Y plane to ensure completion of the vertical pulling motion before the end-effector reaches to the obstacle's projection in the same plane.

Upon full execution of the downward displacement, the OBSTACLE state is activated where the arm remains at the same elevation level ($z^{adapt}(t) = z^{adapt}(t-1)$) until the obstacle is passed. This state maintains continuous safety verification by comparing h^{sh} with $z^{EE,obs}(t)$, where the latter represents the current obstacle height information on top of the end-effector. Since the end-effector tracks previously passed base poses (see Algorithm 1, lines 9–15), the corresponding obstacle height $z^{EE,obs}(t)$ is retrieved from the $\mathbf{Z}^{B,obs}$ at each control cycle as:

$$z^{EE,obs}(t) = z^{B,obs}(t_j), \text{ where } z^{B,obs}(t_j) \in \mathbf{Z}^{B,obs}, \quad (24)$$

with t_j is the time instant at which the base's past position most closely matches the current desired end-effector position on the X - Y plane:

$$t_j = \arg \min_{t'} \|({}_B X_x(t'), {}_B X_y(t')) - ({}_{EE} X_x^{des}(t), {}_{EE} X_y^{des}(t))\|. \quad (25)$$

The retrieved $z^{EE,obs}(t)$ serves as the critical reference for activating the EMERGENCY state while the human is passing beneath the obstacle. In particular, if the vertical distance between the current human head and the obstacle above the end-effector falls below the predefined safety margin (Δ_{safe}):

$${}_{HH} X_z(t) + \Delta_{safe} \geq z^{EE,obs}(t), \quad (26)$$

The system immediately transitions to the EMERGENCY state to mitigate the collision risk.

In the EMERGENCY state, the robot carries out three critical safety actions described below, as long as the condition given in equation (26) holds:

- Base movement is halted by setting the motion enabling parameter $\delta \in \{0, 1\}$ to 0, where $\delta = 1$ in non-emergency states and $\delta = 0$ during INITIALIZATION (see Section 2.3.4 for the details).
- The vertical stiffness is increased by setting $\hat{k}_z = k_z^{max} \in \mathbb{R}_{>0}$.
- A protective motion is started to pull the human down according to:

$$z^{adapt}(t) = \max(h^{min}, z^{adapt}(t-1) - \Delta_{psd}) \quad (27)$$

where $\Delta_{psd} \in \mathbb{R}_{>0}$ defines the per-step descent distance that enables a smooth and comfortable downward motion, and $h^{min} \in \mathbb{R}_{>0}$ sets the lowest possible hand height w.r.t. Σ_B .

Once the safety margin distance (Δ_{safe}) is achieved between the human head and the $z^{EE,obs}(t)$ in the EMERGENCY state, the system transitions back to the OBSTACLE state where the base movement is reactivated by setting $\delta = 1$ and the vertical stiffness is restored to its initial value $\hat{k}_z = k_z^{init} \in \mathbb{R}_{>0}$. The system remains in the OBSTACLE state until the user safely passes the obstacle without any contact, after which it transitions to the OBSTACLE TO FREE state.

For this transition, the human standing height (h^{sh}) is compared with each element in $\mathbf{Z}^{B,obs}$, which stores obstacle height data within a $\zeta^{keep} \in \mathbb{R}_{>0}$ distance long enough to encapsulate the human and robot dyad, as follows:

$$h^{sh} + \Delta_{safe} \leq \min(\mathbf{Z}^{B,obs}). \quad (28)$$

In this state, the robot arm follows a smooth trajectory back to its initial position, governed by the following update rule:

$$z^{adapt}(t) = \min(h^o, z^{adapt}(t-1) + \Delta_{psa}) \quad (29)$$

where $\Delta_{psa} \in \mathbb{R}_{>0}$ is per-step ascent distance facilitating pleasant upward pulling motion. Once full recovery is achieved, the system returns to the FREE state, ensuring that all safety margins are maintained throughout the operation and no further risks are present.

2.3. Control

2.3.1. Cartesian impedance controller. In pHRI scenarios, Cartesian impedance control is a widely used strategy due to its ability to ensure safe interaction between the robot and the human user (Ajoudani et al., 2018b). This control strategy allows the robot to modulate its end-effector dynamics in response to external forces, enabling a compliant and controlled interaction (Ajoudani et al., 2017; Schindlbeck and Haddadin, 2015). This approach relies on

real-time torque sensing and actuation, with the robot arm rigid body model in joint space described by:

$${}_A\boldsymbol{\tau} = \mathbf{M}({}_A\mathbf{q})\ddot{{}_A\mathbf{q}} + \mathbf{C}({}_A\mathbf{q}, \dot{{}_A\mathbf{q}})\dot{{}_A\mathbf{q}} + \mathbf{g}({}_A\mathbf{q}) + {}_A\boldsymbol{\tau}^{ext}, \quad (30)$$

where ${}_A\boldsymbol{\tau} \in \mathbb{R}^n$ is robotic arm joint torques, ${}_A\mathbf{q}, \dot{{}_A\mathbf{q}}, \ddot{{}_A\mathbf{q}} \in \mathbb{R}^n$ are the position, velocity, and acceleration vectors in joint space, respectively, n is the number of joints, $\mathbf{M} \in \mathbb{R}^{n \times n}$ is the joint space inertia matrix, $\mathbf{C} \in \mathbb{R}^{n \times n}$ is the Coriolis and centrifugal matrix, and $\mathbf{g} \in \mathbb{R}^n$ is the gravity term. The external torques (${}_A\boldsymbol{\tau}^{ext} \in \mathbb{R}^n$) generated by the interaction are computed as follows:

$${}_A\boldsymbol{\tau}^{ext} = \mathbf{J}({}_A\mathbf{q})^T \mathbf{F}^{ext} + \boldsymbol{\tau}^{st}, \quad (31)$$

where $\mathbf{J} \in \mathbb{R}^{6 \times n}$ is the robot arm Jacobian matrix, $\mathbf{F}^{ext} \in \mathbb{R}^6$ represents the interaction forces in Cartesian space, and $\boldsymbol{\tau}^{st} \in \mathbb{R}^n$ denotes the secondary task torques projected onto the null-space of $\mathbf{J}$.

In this controller, the desired Cartesian impedance behavior in quasi-static form is achieved by:

$$\mathbf{F}^{ext} = \mathbf{K}^{des}({}_{EE}\hat{\mathbf{x}}^{des} - {}_{EE}\mathbf{x}^{des}) + \mathbf{D}^{des}(\dot{{}_{EE}\hat{\mathbf{x}}}^{des} - \dot{{}_{EE}\mathbf{x}}), \quad (32)$$

where $\mathbf{K}^{des} \in \mathbb{R}^{6 \times 6}$ and $\mathbf{D}^{des} \in \mathbb{R}^{6 \times 6}$ represent the desired stiffness and damping matrices, respectively, ${}_{EE}\hat{\mathbf{x}}^{des}$ and ${}_{EE}\mathbf{x} \in \mathbb{R}^6$ are the desired and actual Cartesian poses of the end-effector, and $\dot{{}_{EE}\hat{\mathbf{x}}}^{des}$ and $\dot{{}_{EE}\mathbf{x}} \in \mathbb{R}^6$ denote their corresponding velocity profiles.

2.3.2. Impedance tuning. The authors in Balatti et al. (2020) introduced a self-tuning impedance controller that enables the robot to precisely follow a desired trajectory along the motion vector while remaining compliant in other directions. This compliance allows the robot to adapt smoothly to unintended disturbances, such as obstacles or unexpected human movements. Building upon this foundation, we propose an enhanced version of the algorithm that allows independent regulation of impedance parameters along three principal axes of the impedance ellipsoid, which is defined at the robot's end-effector frame. These axes correspond to (i) the primary direction of movement ($\mathbf{a}_x \in \mathbb{R}^3$), (ii) the lateral axis lying parallel to the ground and perpendicular to the motion direction ($\mathbf{a}_y \in \mathbb{R}^3$), and (iii) the vertical axis perpendicular to the ground ($\mathbf{a}_z \in \mathbb{R}^3$).

By independently adjusting impedance along these axes, the robot achieves the following capabilities:

- Maintaining high stiffness in the movement direction ($\mathbf{a}_x$), ensuring the robot can effectively guide the human user toward the designated goal while providing a stable pulling force.
- Modulating lateral compliance ($\mathbf{a}_y$), allowing free movement when the users follow the predefined safe path or increasing stiffness when the robot needs to

assist in redirecting them onto the correct trajectory (see Section 2.2.2).

- Providing a neutral stiffness profile along the vertical axis ($\mathbf{a}_z$), giving the user the flexibility to make minor hand position adjustments in response to natural variations in posture and movement. However, in the EMERGENCY case, the stiffness is increased to prevent potential collisions (see Section 2.2.3).

To implement this adaptive impedance behavior, the symmetric and positive definite translational components of the Cartesian stiffness and damping matrices, $\mathbf{K}^{trans} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{D}^{trans} \in \mathbb{R}^{3 \times 3}$, are formulated as follows:

$$\mathbf{K}^{trans} = \mathbf{U}\boldsymbol{\Sigma}_k\mathbf{U}^T, \quad (33)$$

$$\mathbf{D}^{trans} = \mathbf{U}\boldsymbol{\Sigma}_d\mathbf{U}^T, \quad (34)$$

where $\boldsymbol{\Sigma}_k \in \mathbb{R}^{3 \times 3}$ and $\boldsymbol{\Sigma}_d \in \mathbb{R}^{3 \times 3}$ are diagonal matrices containing the desired stiffness and damping parameters along the ellipsoid's axes. The matrix $\mathbf{U} \in \mathbb{R}^{3 \times 3}$ represents an orthonormal basis whose columns correspond to the principal impedance axes ($\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$), defined as follows:

$$\mathbf{U} = \begin{bmatrix} \frac{\mathbf{a}_x}{\|\mathbf{a}_x\|}, & \frac{\mathbf{a}_y}{\|\mathbf{a}_y\|}, & \frac{\mathbf{a}_z}{\|\mathbf{a}_z\|} \end{bmatrix} \quad (35)$$

$$\mathbf{a}_x = {}_{EE}\hat{\mathbf{x}}_{\{1:3\}}^{des}(t) - {}_{EE}\hat{\mathbf{x}}_{\{1:3\}}^{des}(t-1), \quad (36)$$

$$\mathbf{a}_y = [-a_{x,y} \ a_{x,x} \ 0]^T, \quad (37)$$

$$\mathbf{a}_z = \mathbf{a}_x \times \mathbf{a}_y, \quad (38)$$

where ${}_{EE}\hat{\mathbf{x}}_{\{1:3\}}^{des}(t) \in \mathbb{R}^3$ and ${}_{EE}\hat{\mathbf{x}}_{\{1:3\}}^{des}(t-1) \in \mathbb{R}^3$ represent the translational components of the desired Cartesian pose ${}_{EE}\hat{\mathbf{x}}^{des}$ at the current and previous control step, respectively, and $\times$ denotes the cross-product operation.

The diagonal stiffness and damping matrices, $\boldsymbol{\Sigma}_k$ and $\boldsymbol{\Sigma}_d$, are formulated as:

$$\boldsymbol{\Sigma}_k = \text{diag}(k_x, k_y, k_z), \quad \boldsymbol{\Sigma}_d = \text{diag}(d_x, d_y, d_z), \quad (39)$$

where k_x, k_y , and $k_z \in \mathbb{R}_{\geq 0}$ define the desired stiffness values projected onto $\mathbf{U}$ basis axes. The damping coefficients ($d_i \in \mathbb{R}_{\geq 0}$) in modal coordinates are calculated using the relationship $d_i = 2\zeta\sqrt{k_i}$ for $i \in (x, y, z)$ (Albu-Schaffer et al., 2003), and transformed back to the Cartesian coordinated using (34) to ensure stable impedance behavior. Note that k_y and k_z are set equal to $\bar{k}_y$ and $\bar{k}_z$, respectively, which are obtained from the *Horizontal* and *Vertical Pulling* units.

2.3.3. Cartesian admittance controller. In order to control mobile platforms, velocity-based low-level controllers are commonly employed. These controllers are designed with high gains that help to eliminate the effects of disturbances, which mainly arise from dynamic couplings between the

arm and the mobile base. In addition, they compensate for external forces and the mobile platform's own dynamic behavior.

In this study, we utilize a Cartesian admittance controller to map the high-level torques, which are calculated by the *Virtual Torques Shaping* unit, into the desired velocities for the mobile base. For the mobile bases with omnidirectional wheels, the number of DoFs (m) is 3, since they can move along the X and Y directions as well as rotation around the Z axis (yaw), without being constrained by non-holonomic limitations. The dynamics of these mobile platforms, modeled virtual joints (${}_B\mathbf{q} \in \mathbb{R}^m$), can be described by the following equation:

$$\mathbf{M}^{adm}{}_B\ddot{\mathbf{q}}^{des} + \mathbf{D}^{adm}{}_B\dot{\mathbf{q}}^{des} = {}_B\boldsymbol{\tau}^{vir} + {}_B\boldsymbol{\tau}^{ext}, \quad (40)$$

where $\mathbf{M}^{adm} \in \mathbb{R}^{m \times m}$ and $\mathbf{D}^{adm} \in \mathbb{R}^{m \times m}$ are the diagonal positive definite virtual inertia and virtual damping, ${}_B\dot{\mathbf{q}}^{des}$, ${}_B\ddot{\mathbf{q}}^{des} \in \mathbb{R}^m$ are the desired velocity sent to the mobile platform and the acceleration, ${}_B\boldsymbol{\tau}^{vir} \in \mathbb{R}^m$ and ${}_B\boldsymbol{\tau}^{ext} \in \mathbb{R}^m$ correspond to the virtual and external torques, respectively.

The ${}_B\boldsymbol{\tau}^{vir}$ is composed of the linear forces along the X and Y axis (f_x and $f_y \in \mathbb{R}$), and the rotational torque around the Z axis ($\mu_z \in \mathbb{R}$). It can be expressed as:

$${}_B\boldsymbol{\tau}^{vir} = [f_x, f_y, \mu_z]^T. \quad (41)$$

The desired velocity (${}_B\dot{\mathbf{q}}^{des}$) can be computed by reformulating the acceleration term as ${}_B\ddot{\mathbf{q}}^{des}(t) = t_s^{-1}({}_B\dot{\mathbf{q}}^{des}(t) - {}_B\dot{\mathbf{q}}^{des}(t-1))$ and substituting this expression into the system dynamics:

$${}_B\dot{\mathbf{q}}^{des}(t) = (t_s^{-1}\mathbf{M}^{adm} + \mathbf{D}^{adm})^{-1}({}_B\boldsymbol{\tau}^{vir}(t) + {}_B\boldsymbol{\tau}^{ext}(t) + t_s^{-1}\mathbf{M}^{adm}{}_B\dot{\mathbf{q}}^{des}(t-1)). \quad (42)$$

2.3.4. Virtual torques shaping. This unit calculates the virtual torques input (${}_B\boldsymbol{\tau}^{vir}$) for the *Cartesian Admittance Controller* based on the base's current frame (Σ_B), the next target waypoint of the base (Σ_{Bdes}), the maximum allowable wrench exerted on the human operator ($\mathcal{F}^{des} \in \mathbb{R}^3$), the gain (α) that dynamically adjusts in response to the human's movement delay (see Section 2.2.2) and the gain (δ) that mitigates collision with the upper body (see Section 2.2.3). The virtual torques are computed for each component as follows:

$${}_B\boldsymbol{\tau}^{vir}(f_x) = \vec{\mathbf{w}}_x \mathcal{F}_x^{des} \alpha \delta, \quad (43)$$

$${}_B\boldsymbol{\tau}^{vir}(f_y) = \vec{\mathbf{w}}_y \mathcal{F}_y^{des} \alpha \delta, \quad (44)$$

$${}_B\boldsymbol{\tau}^{vir}(\mu_z) = \vec{\mathbf{w}}_\theta \mathcal{F}_\theta^{des} \alpha \delta, \quad (45)$$

where $\vec{\mathbf{w}} = [x, y, \theta]^T \in \mathbb{R}^3$ represents the vector between Σ_B and Σ_{Bdes} , defined as:

$$|\vec{\mathbf{w}}| = \sqrt{{}^B_{Bdes}\mathbf{X}_x^2 + {}^B_{Bdes}\mathbf{X}_y^2}, \quad (46)$$

$$\vec{\mathbf{w}}_x = {}^B_{Bdes}\mathbf{X}_x / |\vec{\mathbf{w}}|, \quad (47)$$

$$\vec{\mathbf{w}}_y = {}^B_{Bdes}\mathbf{X}_y / |\vec{\mathbf{w}}|, \quad (48)$$

$$\vec{\mathbf{w}}_\theta = \text{atan2}(-\vec{\mathbf{w}}_y, \vec{\mathbf{w}}_x). \quad (49)$$

3. System overview

In this study, we employed the MOCA mobile manipulator as the robotic assistant (Wu et al., 2019). It consists of an Omnidirectional Robotnik SUMMIT-XL STEEL mobile platform with a lightweight, torque-controlled 7-DoF Franka Emika Panda arm mounted on top. Additionally, an underactuated Pisa/IIT SoftHand was attached to the arm's end-effector, allowing humans to hold it during guidance. The mobile base is equipped with two 2D SICK Microscan LiDAR sensors, positioned at the front and back of the robot, 30 cm above the ground, providing a 360-degree sensing range. These sensors detect human legs and generate collision-free waypoints for the base toward the desired target location.

For experiments involving upper-body obstacle obstruction, additional sensors were integrated onto the robot. A 3D LiDAR (Ouster OS0-32, $\pm 45^\circ$ field of view) was mounted at the back of the base (see Figure 1) to capture the area above the robot. Additionally, an RGB-D camera (Intel RealSense D435i) was attached to a pole on the base to track the human head in real time.

The robot's software architecture is built on the robotics middleware Robot Operating System (ROS), utilizing C++ as the client library. Each unit detailed in Section 2 is implemented as a ROS node, and data communication between these units takes place through ROS messages on the ROS topics illustrated in Figure 2, employing the publisher/subscriber design pattern. All units run continuously, with their processes updating at specific rates. The *Human Guidance* unit operates at 1000 Hz, while collision-free waypoints are generated at 20 Hz by the *move_base* package using 2D LiDAR data. Both the *Horizontal Pulling* and *Vertical Pulling* units run at 200 Hz. On the other hand, the *Cartesian Impedance Controller* and *Cartesian Admittance Controller* update at 1000 Hz and 300 Hz, respectively. For human tracking, the *Legs Tracker* processes data at 40 Hz and the *Head Tracker* at 15 Hz. Finally, the 3D LiDAR provides point cloud data for detecting upper-body obstacles at 10 Hz.

4. Experiments and results

To validate the effectiveness of the proposed approach, we conducted a series of experiments where sighted participants who were blindfolded were guided by the robotic assistant toward a target position while navigating around obstacles.

The first set of experiments focused on evaluating the performance of the *Horizontal Pulling* strategy. In these trials, the goal was to assess how effectively the robot could generate adaptive pulling motion to keep the human aligned with a collision-free path while avoiding both static and dynamic obstacles detectable by the robot's 2D LiDAR sensors, such as walls and walking pedestrians.

In contrast, the second set of experiments also introduced obstacles positioned at various heights, posing a risk to the upper body of the guided individual. To handle these more complex scenarios, the activation of the *Vertical Pulling* strategy is required, as well as the *Horizontal Pulling*. By incorporating this additional layer of obstacle avoidance, we evaluated the system's capability to ensure full-body safety while maintaining a smooth guidance experience.

A video of these experiments, along with the proof-of-concept demonstrations, can be found at: https://youtu.be/_ZubNEGhSGQ.

4.1. Experiments without upper-body obstacles

4.1.1. User study with predefined path. These experiments were designed to verify the effectiveness of the *Horizontal Pulling* strategy by comparing the results obtained with and without its activation. To this end, the participants were asked to be guided by holding the robotic hand at the end-effector of our robotic assistant along a predefined trajectory while blindfolded in order to simulate a visual impairment. The trials were randomized and conducted in a controlled laboratory environment with 12 healthy subjects (6 males and 6 females; age: 28.9 ± 2.7 years; height: 172.4 ± 8.1 cm). Before the experiment, a familiarization trial was conducted without the blindfold on a simpler trajectory to prevent memorization of the main trajectory. The entire experimental procedure was carried out at the Human-Robot Interfaces and Interaction (HRII) Lab, Istituto Italiano di Tecnologia (IIT), in accordance with the Declaration of Helsinki. The protocol was approved by the ethics committee Azienda Sanitaria Locale (ASL) Genovese N.3 (Protocol IIT HRII ERGOLEAN 156/2020).

4.1.1.1. Controller parameters. In order to allow subjects the freedom of lateral motion in the baseline trials (i.e., without *Horizontal Pulling*), the lateral stiffness of the robot k_y was set to 0 N/m. In contrast, as illustrated in Figure 3, the *Horizontal Pulling* unit maintains lateral stiffness near zero for small deviations—ensuring a fair comparison with the baseline—but increases it sharply as deviations grow.

For both conditions, the stiffness in the primary direction of motion was set to $k_x = 1000$ N/m, providing firm guidance along the intended path. The vertical stiffness was set to $k_z = 500$ N/m, enabling users to make minor adjustments to the hand's height. These values were chosen within the typical range of human arm stiffness to maintain a natural and comfortable interaction (Ajoudani et al., 2018a). Additionally, the desired pulling wrench ($\mathcal{F}^{des}$) was set to 80 N for the X and Y axes, and 50 Nm about the Z -axis, while the mobile base admittance control parameters were set to $\mathbf{M}^{adm} = \text{diag}\{32.5, 32.5, 10.8\}$, and $\mathbf{D}^{adm} = \text{diag}\{325, 325, 108.3\}$, to provide a controlled and safe velocity for visually impaired users. The distances used to determine the desired poses of the mobile base, end-effector, and the human were set to $D_1 = 0.15$ m, $D_2 = 0.8$ m, and $D_3 = 0.4$ m, respectively. Moreover, h° was selected to 1.1 m to provide a comfortable grasping height for the user. Lastly, the saturation distance d^{stop} was set to -0.4 m, meaning the robot base should stop if the human lags behind by more than this distance.

4.1.1.2. Results. Figure 5 presents the experimental results for a single participant who performed the experiment as described in Section 4.1.1. The figure displays the trajectories followed by the human-robot dyad on the X - Y plane, with (a) and without (b) using the *Horizontal Pulling* strategy. To better visualize the trials, Figures 5(a) and (b) include the target waypoints of the base (${}_B\mathbf{X}^{targets}$) as turquoise, the mobile base path that tracks the waypoints (${}_B\mathbf{X}$) as a black line, human leg path (${}_{HL}\mathbf{X}$) as a green line and the actual end-effector path (${}_{EE}\mathbf{X}$) as a red line, and the robot's base footprint as a shaded gray area. The desired end-effector (${}_{EE}\mathbf{X}^{des}$) and human (${}_{HL}\mathbf{X}^{des}$) poses are not plotted, as they correspond to ${}_B\mathbf{X}$ shifted back in time, as explained in Algorithm 1, making them not visible. In addition to these, Figure 5(a) illustrates the ${}_{EE}\hat{\mathbf{X}}^{des}$, the adjusted end-effector desired pose according to *Horizontal Pulling* strategy, as a dashed purple line.

During the trial with the *Horizontal Pulling* active, it can be seen that ${}_{EE}\hat{\mathbf{X}}^{des}$ compensates for the human deviations from the desired trajectory (see Figure 5(a)). For example, when the human deviated to the robot's right in the focused portion of the plot (point A), the translation of y^{adapt} applied to ${}_{EE}\mathbf{X}^{des}$ shifts the robot's end-effector desired pose (${}_{EE}\hat{\mathbf{X}}^{des}$) to the robot's left (point B). This adjustment helps pull the human back toward the desired collision-free path. As a result, the human leg pose ${}_{HL}\mathbf{X}$ never extends beyond the robot's base footprint, indicated by the shaded gray area. Conversely, in the absence of *Horizontal Pulling* strategy (see Figure 5(b)), the human's lateral deviations become significantly larger, frequently extending beyond the robot's base footprint. Specifically, the participant exceeds this boundary for 22.84% of the trial duration, which could increase the likelihood of unintended collisions.

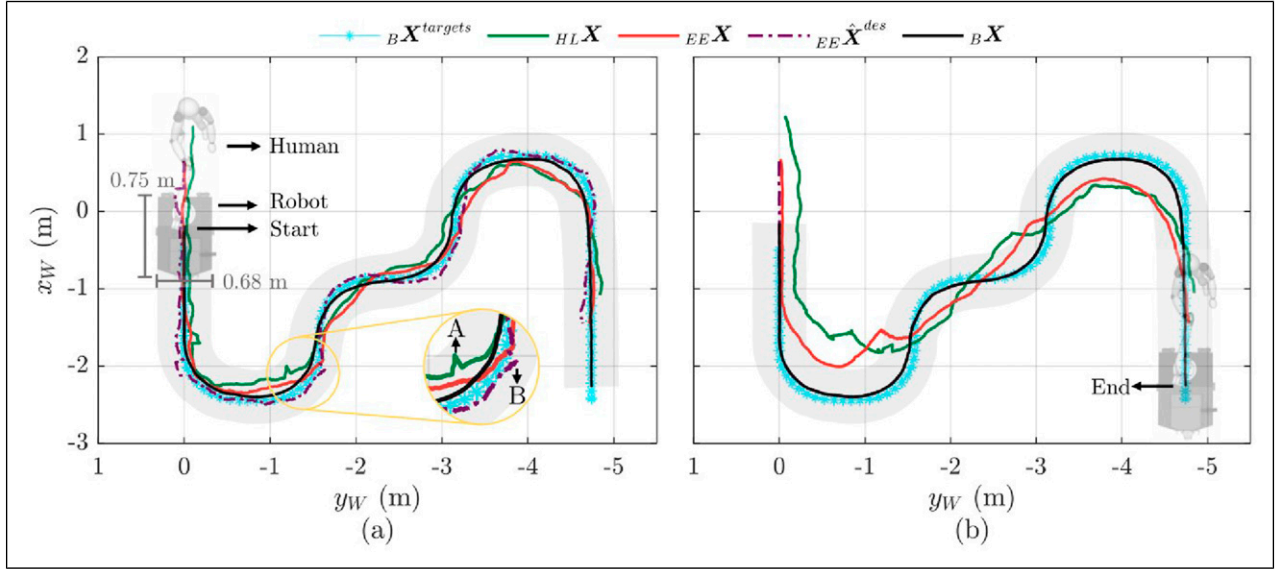


Figure 5. Results of the trials of a single participant. The plots illustrate the trajectories followed by the human–robot dyad on the X – Y plane along a predefined path, shown with (a) *Horizontal Pulling* enabled and (b) disabled. The turquoise line with (*) represents the target waypoints of the base (${}_B\mathbf{X}^{targets}$), the green, red, and black lines depict the human leg path (${}_{HL}\mathbf{X}$), actual end-effector path (${}_{EE}\mathbf{X}$), and mobile base path that tracks the waypoints (${}_B\mathbf{X}$), respectively; the dashed purple line indicates the adjusted end-effector desired path (${}_{EE}\hat{\mathbf{X}}^{des}$) according to the *Horizontal Pulling* strategy, and the shaded gray area shows the robot’s base footprint.

Figure 6 presents the data collected from the example trial where *Horizontal Pulling* is active, with (a) illustrating the saturated pulling vector ($\vec{p}^{sat}$), (b) the corresponding adjustments in lateral stiffness ($\hat{k}_y$), and (c) the resulting changes in base virtual torque gain (α). As it can be seen, $\hat{k}_y$ varies according to $\vec{p}_y^{sat}$ as defined in equation (14). Specifically, when the participant deviates laterally from the intended trajectory, $\hat{k}_y$ increases to generate a stiffer pull to guide the human back toward the desired path. On the other hand, the gain α is regulated by $\vec{p}_x^{sat}$ according to equations (15) and (16), decreasing below 1 whenever $\vec{p}_x^{sat}$ becomes negative. This reduction in α leads to a decrease in the base velocity of the robot, effectively slowing down the system when the participant lags behind their expected position (Σ_{HL}^{des}). The small fluctuations in $\vec{p}_x^{sat}$ arise from the natural variations in human footsteps. As discussed in Section 2.1.1, the human leg pose is estimated by averaging the centroid positions of both legs, which introduces slight oscillations in the recorded data.

Figure 7 shows the results obtained from all participants during the user study: (a) and (b) display the trajectories followed by the human–robot dyads on the X – Y plane with and without *Horizontal Pulling*, respectively; (c) and (d) show bar plots of the average absolute deviations in $|\vec{p}_x|$ and $|\vec{p}_y|$, respectively; and (e) presents the deviation trends of $|\vec{p}_y|$ over normalized time. As seen in Figures 7(a) and (b), the proposed adaptive guidance approach effectively kept all the participants within safe boundaries, defined as the area

where the robot base moved without colliding with the environment, whereas in baseline trials, the participants frequently strayed from the collision-free path, increasing the risk of unsafe interactions with the environment. The effectiveness of the *Horizontal Pulling* strategy in mitigating unintended lateral deviations is also evident in Figure 7(d), where a statistically significant decrease in mean $|\vec{p}_y|$ ($p < 0.001$, sign-test) is observed with the proposed method. Moreover, Figure 7(e) illustrates that this lateral deviation remains consistently higher across almost all segments of the experiment in the baseline condition compared to the proposed strategy. On the other hand, the statistically significant reduction in $|\vec{p}_x|$ (see Figure 7(c)) shows that *Horizontal Pulling* strategy is not only effective in reducing lateral deviations but also in maintaining the intended following distance between human and robot by modulating the α .

4.1.2. User study with dynamic obstacles. In addition to the previous user study, we conducted a second user study focusing on the system’s ability to generate collision-free trajectories in real time while navigating in environments containing dynamic obstacles, as detailed in Section 2.2.1. Unlike the prior experiment, which primarily evaluated the *Horizontal Pulling* strategy along a predefined path, this study further evaluates how effectively the system detects moving obstacles and adapts its trajectory dynamically to prevent collisions while guiding the user toward a target.

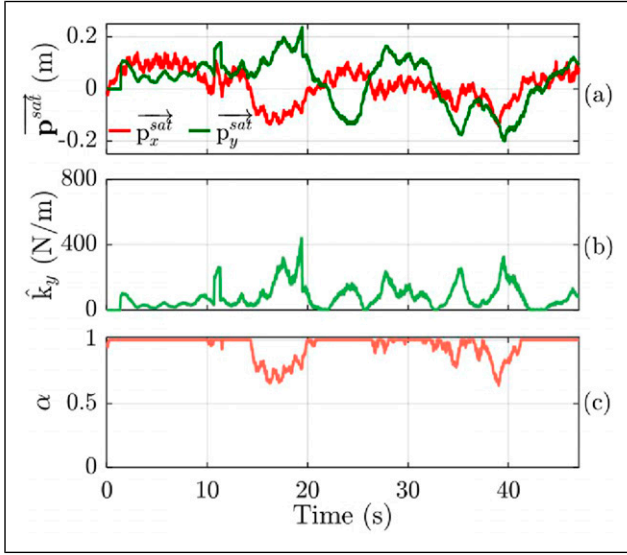


Figure 6. Effect of $\vec{p}^{sat}$ components: $\vec{p}_x^{sat}$ adjusts base torque gain α , while $\vec{p}_y^{sat}$ modulates lateral stiffness $\hat{k}_y$.

For the experiments that took place in a controlled laboratory environment similar to the prior one, two different scenarios were designed where six blindfolded participants (3 males and 3 females; age: 29.8 ± 1.9 years; height: 167.1 ± 8.5 cm) were guided toward a goal located 4.7 m directly behind the robot. In the first scenario, a single dynamic obstacle approached critically close to the robot's estimated path during the course of the movement. On the other hand, the second scenario featured an additional dynamic obstacle appearing after the first, positioned between the initial obstacle and the robot, to evaluate the robot's ability to continually replan as the obstacles are progressively detected. Given that pedestrians are the most common dynamic obstacles encountered in everyday environments, human walkers were utilized as dynamic obstacles in these experiments. Each participant completed both scenarios, following the same ethical protocol and using the same controller parameters as in the previous experiment.

4.1.2.1. Results. Figure 8 depicts the results from a single participant in the two previously described scenarios involving (a) one dynamic obstacle and (b) two dynamic obstacles. The figure shows the trajectories of both the human-robot dyad and the dynamic obstacles on the X - Y plane. The trajectories of the dynamic obstacles were recorded using the OptiTrack motion capture system; however, this information was used solely for visualization and was not integrated into our guidance approach, which relies on 2D LiDAR sensors mounted at the base for obstacle detection. The trajectories of each obstacle, the human leg, and the mobile base are plotted as continuous lines, with colors varying over time according to the provided colormap.

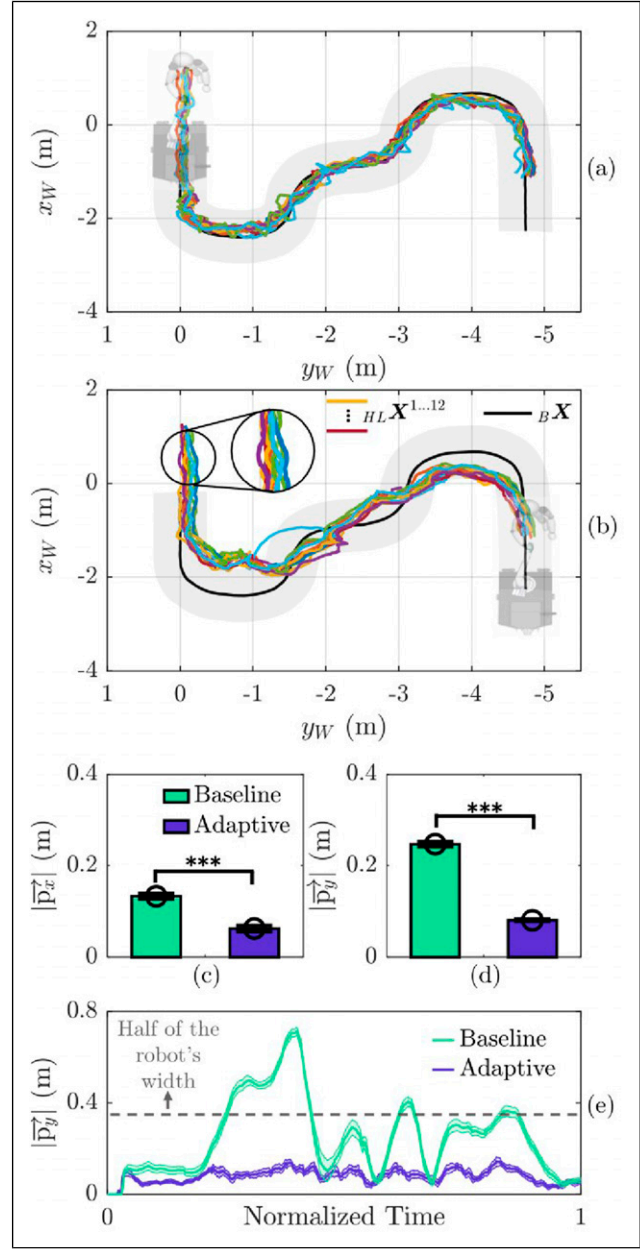


Figure 7. The results from the user study across all participants, with 12 colored lines representing the individual leg positions ($_{HL}\mathbf{X}$) of each participant, and the shaded gray area indicating the robot's base footprint. The graphs show (a) the trajectory of the human-robot dyad on the X - Y plane along a predefined path with adaptive Horizontal Pulling enabled, (b) with it disabled, (c) the average absolute deviation in $|\vec{p}_x^{sat}|$, (d) the average absolute deviation in $|\vec{p}_y^{sat}|$, and (e) the deviation trends of $|\vec{p}_y^{sat}|$ over normalized time. Additionally, the results of the sign-test comparing the Horizontal Pulling strategy with baseline trials are included, with *** denoting $p < 0.001$.

As can be seen, the human-robot dyad successfully reached the desired target in both trials while avoiding the dynamic obstacles. Thanks to the Human Guidance unit (see Section 2.2.1), the proposed system continuously detected

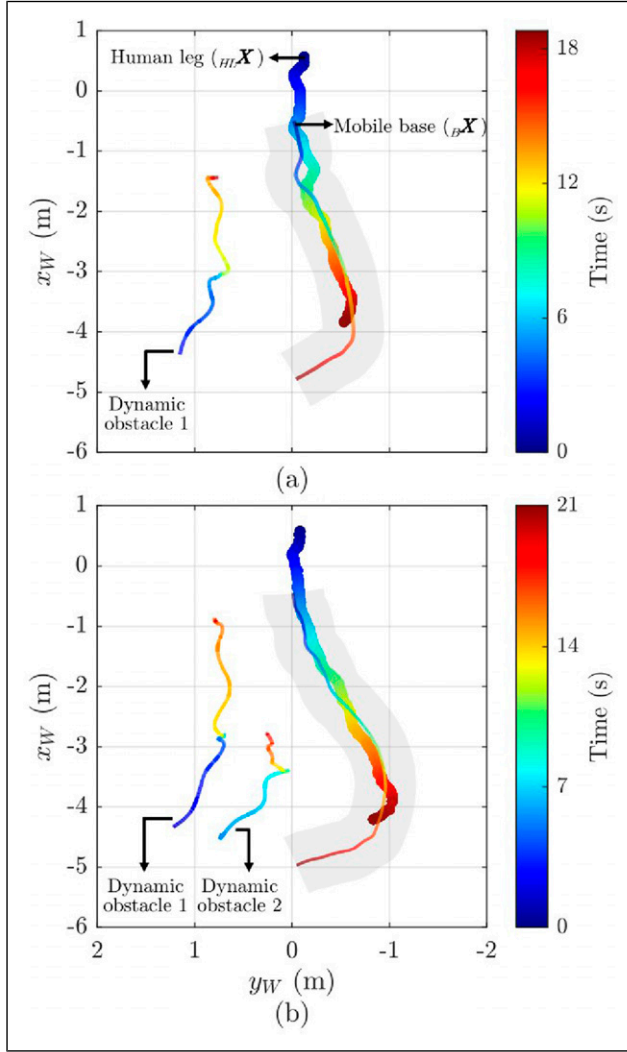


Figure 8. Results of the trials of a single participant in the dynamic obstacle experiments. The plots show the trajectories of the human–robot dyad and dynamic obstacles on the X – Y plane when (a) an obstacle is present and (b) two obstacles are present. The trajectories of the obstacles, the human leg, and the mobile base are drawn as continuous lines whose colors vary over time according to the provided colormap, and the shaded gray area indicates the robot's base footprint.

the dynamic obstacles within the environment and generated collision-free paths by continuously updating the local plan. The comparison of the trajectories in each scenario reveals that the robot adjusted its path whenever an obstacle moved closer, effectively adapting to changing conditions in real time. Moreover, when an obstacle was detected, the system continued to guide the participant without producing abrupt or erratic motions, ensuring a smooth guidance experience.

Figure 9 shows the means and standard errors of the total traversed path length of the mobile base ($\|_B X\|$) across both scenarios for all participants, along with the results of the sign-test. As expected, the path length in the second scenario was significantly longer than in the first, reflecting the need

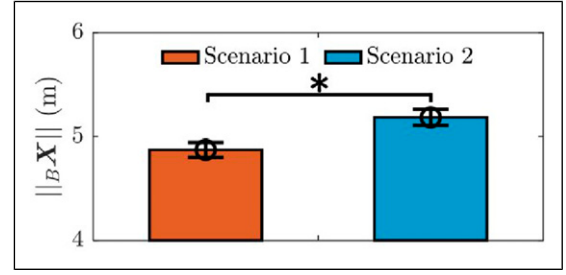


Figure 9. The means and standard errors of the total traversed path length of the mobile base ($\|_B X\|$) across both scenarios for all participants. The result of the sign-test is also shown, with * indicating $p < 0.05$.

to safely navigate around the additional obstacle. Importantly, due to the adaptive replanning capability of the proposed system, all participants successfully reached the goal position in both scenarios without any collisions.

4.1.3. Practical real-world test. In order to complement the controlled laboratory experiments, we conducted a practical real-world test to evaluate the system's performance in providing safe and efficient guidance within an unstructured environment. This test can also be considered as a proof-of-concept, demonstrating that despite the user's unfamiliarity with the environment, the system can leverage a provided global map to generate collision-free trajectories in real time and successfully guide the user to the desired target.

Figure 10 illustrates the generated collision-free trajectories for the human–robot dyad while navigating in a narrow corridor, using the same color coding as in Figure 5. Additionally, the plot features an environment map that provides a visual representation of the obstacles detected by the system, along with markers indicating the locations where snapshots of the pivotal stages (points A, B, C, and D) were captured during the trial. These stages verify the successful integration of human–robot collaboration (HRC) system components, allowing the robot to guide the human through complex environments with sharp 90-degree turns while proactively avoiding obstacles.

4.2. Experiments with upper-body obstacles

4.2.1. User study scenario. These experiments aimed to assess the overall performance of the proposed approach that provides a three-dimensional navigation assistance (i.e., activating *Horizontal Pulling* and *Vertical Pulling*) through both quantitative metrics and qualitative user feedback. Similar to the user studies described in Sections 4.1.1 and 4.1.2, blindfolded participants were guided by an anthropomorphic robotic hand attached to the end-effector along a predetermined path featuring multiple 90-degree turns. Additionally, in these experiments, two obstacles with varying heights (see Figure 11) were placed at different locations along the path to simulate the risk of upper-body

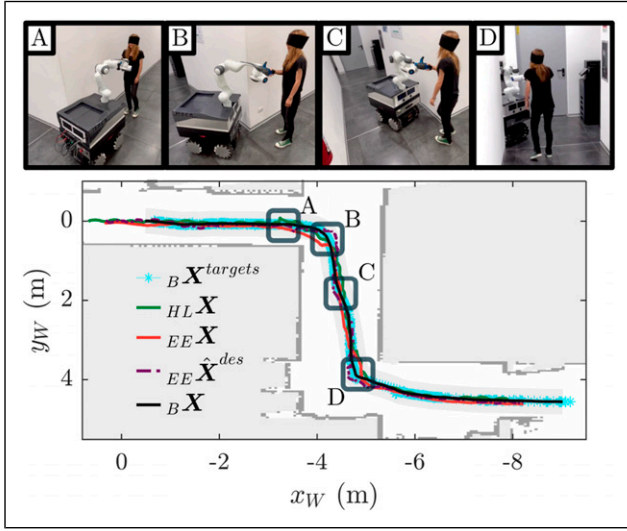


Figure 10. The result of *Practical Real-world Test* where *Human Guidance* unit continuously generates collision-free path $B X^{targets}$ and updates next desired base position while localizing the robot within the map of the environment. Screenshots of four significant moments during the interaction are shown in (a)–(d).

collisions. Ten participants (5 female, 5 male; age: 29.1 ± 3.2 years; height: 169.9 ± 8.7 cm)² took part in the study. Each participant completed a familiarization trial on a shorter trajectory featuring a single 90-degree turn and one upper-body obstacle prior to the main experiment. At the end of the experiment, participants completed the System Usability Scale (SUS) questionnaire (Brooke, 1995), a standardized 10-item survey based on a 5-point Likert scale, to qualitatively assess user experience with the proposed assistive guidance. Note that the whole experimental procedure was in accordance with the same ethical standards as in Section 4.1.1.

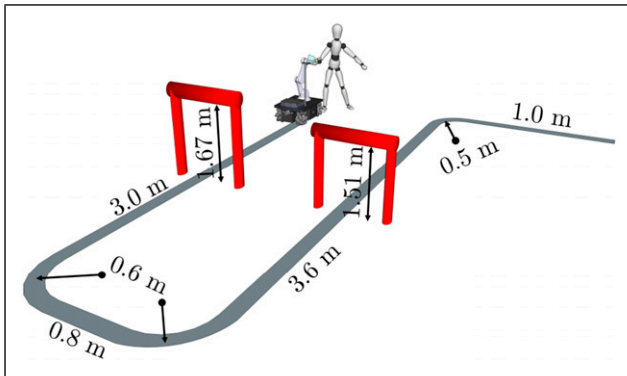


Figure 11. Illustration of the experiment with upper-body obstacles, where the shape of the path and the locations of two obstacles are shown. These experiments were designed to evaluate the overall performance of the proposed approach for three-dimensional navigation.

4.2.2. Controller parameters. The Cartesian stiffness defined in Σ_{EE} was independently controlled along each axis: (i) for X -axis motion, a constant high stiffness of $k_x = 1000$ N/m maintained trajectory precision; (ii) Y -axis stiffness, $k_y = \hat{k}_y$, varied nonlinearly to adjust lateral compliance (see Figure 3); and (iii) Z -axis stiffness $k_z = \hat{k}_z = 400$ N/m maintained nominal stiffness but increased to 800 N/m in EMERGENCY state for exerting stronger pulling force toward desired vertical position. The same pulling wrench ($\mathcal{F}^{des}$), the mobile base admittance control parameters (M^{adm} and D^{adm}), and the parameters for determining the desired poses of the HRC system (D_1 , D_2 , D_3 , and h°) were used as in the previous experiments to ensure a comfortable motion for blindfolded users.

The span of the region of interest (ROI) used in *Vertical Pulling* unit was defined by the following boundaries relative to the Σ_B frame. The X -axis limits were set to $\epsilon_x^{min} = -0.6$ m and $\epsilon_x^{max} = -0.38$ m, where ϵ_x^{max} aligns with the rear edge of the robot base, while ϵ_x^{min} extends further behind the base. This allowed us to identify potential obstacles in the robot's backward path before they got critically close. The symmetric Y -axis boundaries ($\epsilon_y^{min} = -0.34$ m, $\epsilon_y^{max} = 0.34$ m) corresponded to the full lateral extent of the base. Vertical detection was constrained between $\epsilon_z^{min} = 1.4$ and $\epsilon_z^{max} = 2.5$ m; objects located above ϵ_z^{max} were ignored, as they pose no hazard to the user. This constrained ROI configuration significantly reduced the size of the 3D point cloud compared to full-space monitoring, thereby lowering the computational load.

The obstacle history keeping distance was set to $\zeta^{keep} = 1.3$ m, deliberately exceeding the base-to-human-leg separation, to ensure that the stored obstacle data always included the human during vertical pulling. The safety margin $\Delta_{safe} = 0.05$ m was defined as the minimum permissible distance between the human head and the obstacle above. The controlled lowering motion in FREE TO OBSTACLE state executed a pull of $\Delta_{down} = 0.45$ m, with a required travel distance of $\zeta^{req} = 0.75$ m, ensuring the downward motion was completed before the end-effector reaches a point below the obstacle's location. With the control loop operating at 200 Hz, the EMERGENCY state used a protective per-step downward motion at a rate of $\Delta_{psd} = 0.0005$ m/cycle (10 cm/s) until the human head height became lower than the $z^{EE,obs}$. Nevertheless, if the hand reached the lowest permissible height ($h^{min} = 0.6$ m), any additional downward motion was stopped to avoid exceeding the robotic arm's workspace limits. Upon successful obstacle avoidance, the arm initiated recovery with an ascent rate of $\Delta_{psa} = 0.00125$ m/cycle (25 cm/s) to return to the initial height. These velocities were selected to ensure user comfort while maintaining safety, corresponding to smooth yet effective motion profiles.

4.2.3. Results. Figure 12 presents experimental results from two participants who were guided using three-dimensional

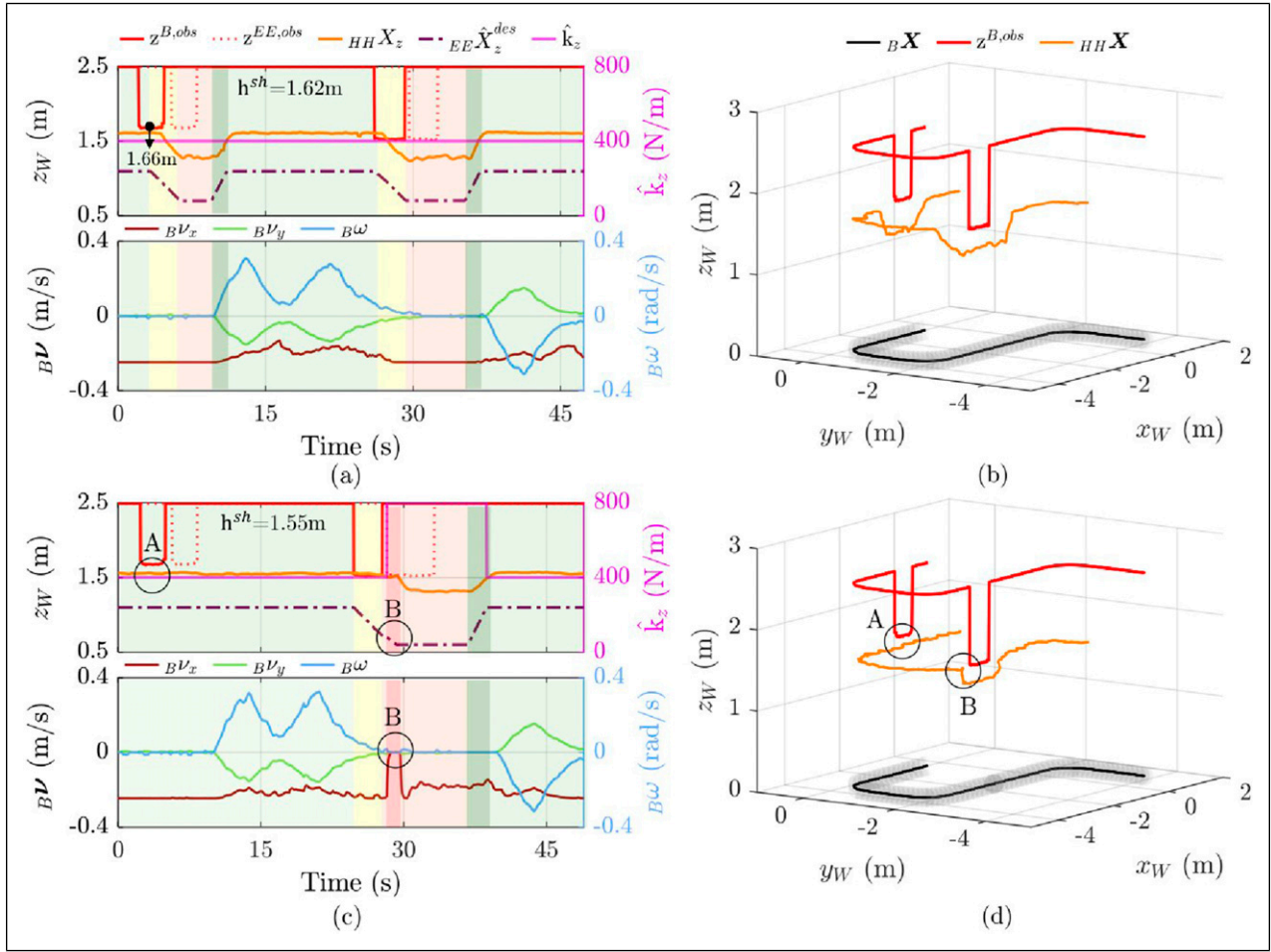


Figure 12. Experimental results from two participants experiencing both vertical and horizontal assistance, with (a) and (b) showing data from the first participant and (c) and (d) from the second. The light green, yellow, orange, red, and dark green areas correspond to the FREE, FREE TO OBSTACLE, OBSTACLE, EMERGENCY, and OBSTACLE TO FREE states, respectively. The participants' standing heights (h^{sh}), calculated upon completion of INITIALIZATION, are annotated in the top plots of sub-figures (a) and (c).

assistance, following the scenario described in Section 4.2.1. The subplots in Figures 12(a) and (b) display data from the first participant, while Figures 12(c) and (d) correspond to the second. To analyze the *Vertical Pulling* unit in detail, the Z-axis positions w.r.t. Σ_W frame (i.e., heights) of the human head (HHX_z), desired end-effector ($EE\hat{X}_z^{des}$), obstacle above the base ($z^{B,obs}$) and the end-effector ($z^{EE,obs}$) as well as the vertical stiffness ($\hat{k}_z$) and the base's linear (BV_x and BV_y) yaw ($B\omega$) velocities are reported.

The colored regions in the plots indicate the current state of the *Vertical Pulling* unit. The 3D plots in (b) and (d) visualize the robot base's trajectory alongside the human head and obstacle positions in Cartesian space, providing a clearer understanding.

For the first participant, the FREE state remained active (light green interval in Figure 12(a)) until the first obstacle entered the ROI of the *Vertical Pulling* unit. In particular, when the $z^{B,obs}(t)$ became critically close to the human

standing height (i.e., $z^{B,obs}(t) < h^{sh} + \Delta_{safe}$) the system transitioned to the FREE TO OBSTACLE state. Over the course of this state, the robot arm followed the downward pulling trajectory defined in equation (22) as shown in the yellow interval. As soon as the planned pulling-down motion was completed, the system transitioned into the OBSTACLE state (orange interval). The end-effector maintained the same vertical position (see dashed purple line) until the base had traveled far enough to allow the human to pass beneath the obstacle. Once the condition $z_i^{B,obs} \geq h^{sh} + \Delta_{safe}$ was satisfied for all $z_i^{B,obs} \in Z_{obs}^B$, the OBSTACLE TO FREE state was initiated. During this state, the end-effector smoothly ascended, as shown in the dark green highlighted region. When the initial Z-axis position was restored, the system re-entered the FREE state. A similar sequence of state transitions occurred for a subsequent obstacle, where the proposed planning unit provided a safe passage for the

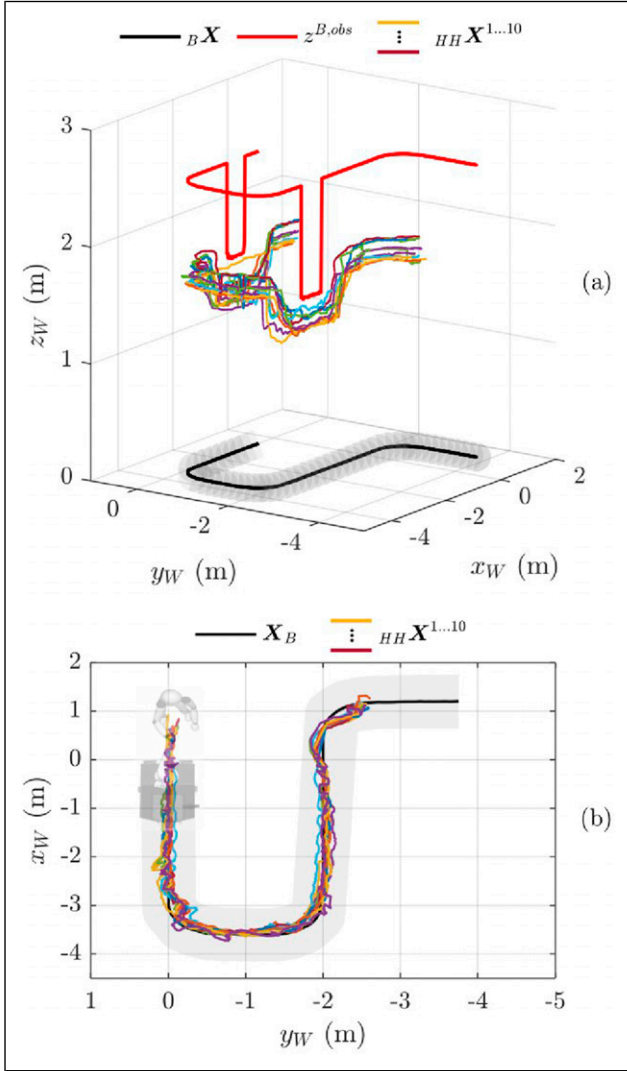


Figure 13. The results from the trials involving all participants, under the condition where both the *Horizontal* and *Vertical Pulling* units were enabled. The solid lines—excluding the red (closest obstacle location) and black (base path)—represent the head positions in 3D Cartesian space (a), with their corresponding leg positions projected onto the X - Y plane shown in (b).

user, although it was closer to the ground compared to the first one.

In the case of the second participant, the *FREE* state was remained active throughout passage under the first obstacle (see black circle marked as A), due to the fact that the inequality $z^{B,obs}(t) \geq h^{sh} + \Delta_{safe}$ always held. This demonstrates the proposed approach's ability to assess collision risk and opt against vertical adjustment when it is unnecessary. For the second obstacle, despite the end-effector having already descended during the *OBSTACLE TO FREE* state, no notable reduction in human head height was observed (see orange line within the yellow interval). As a result, the *EMERGENCY* state was triggered because the system predicted a high risk of contact in the near future.

This high risk is noticeable when examining the start of the red interval, as $z^{EE,obs} \approx h^{sh}$ at that point. The corresponding spatial representation in Figure 12(d) clearly shows that the user was still at a safe distance from the obstacle when the system entered the *EMERGENCY* state, demonstrating the effectiveness of the proposed approach in proactively ensuring safety.

Figure 13 illustrates the positions of the human head (a) and leg (b) in the Σ_W frame across all subjects. The black line represents the trajectory of the mobile platform, while the red line marks the closest obstacle height on top of the base. The results show that each participant was safely guided to the target location, with no collisions. Furthermore, the lateral deviation from the desired path remained within the safe zone (light grey area, corresponding to the width of the robot) throughout all trials.

The outcomes of the SUS questionnaire are reported in Table 1 where the performance of the system in the aspects of effectiveness, efficiency, and overall ease of use is evaluated. This table presents the mean raw scores for each question and the average percentile ranking for all participants, which is calculated by normalizing the raw scores and converting them into percentile rankings (ranging from 0 to 100). The proposed HRC system achieved an average percentile ranking of 93.50 ± 6.03 , indicating an “excellent” level of acceptance on the SUS scale (Brooke, 1995). Analyzing the individual questions, the high scores on the positive statements (odd numbered ones) confirmed that the system is easy to adapt to and use, and participants felt confident while using it, even though they were unable to see the environment. Although all scores were close to the maximum ranking of 5, the lowest score, 4.3, indicated an unwillingness to use the system very frequently. This may stem from the discomfort experienced by individuals without visual impairments, who were blindfolded while using the system. The scores for the negative statements (even numbered ones) were close to the lowest possible ranking of 1, implying that the system is not complex, consistent, and accommodating.

5. Discussion and conclusion

In this study, we introduced a novel 3D human guidance approach designed to ensure the full-body safety of visually impaired individuals as they navigate in unstructured environments with obstacles at different levels. Our approach advances beyond traditional cane-based solutions and leash-guided robots by leveraging the loco-manipulation capabilities of a mobile manipulator. The proposed system not only generates collision-free trajectories for the human-robot dyad but also proactively pulls the human back to a safe path upon detecting potential collision risks. Furthermore, in response to the identified risk level, the stiffness of the robotic arm along each axis, as well as the velocity of the

Table 1. System Usability Scale (SUS) questionnaire results.

ID	Statement	Raw score
Q1	I think that I would like to use this system frequently.	4.30 $\pm$ 0.67
Q2	I found the system unnecessarily complex.	1.20 $\pm$ 0.42
Q3	I thought the system was easy to use.	5.00 $\pm$ 0.00
Q4	I think that I would need the support of a technical person to be able to use this system.	1.60 $\pm$ 0.84
Q5	I found the various functions in this system were well integrated.	4.70 $\pm$ 0.48
Q6	I thought there was too much inconsistency in this system.	1.10 $\pm$ 0.32
Q7	I would imagine that most people would learn to use this system very quickly.	4.90 $\pm$ 0.32
Q8	I found the system very cumbersome to use.	1.20 $\pm$ 0.42
Q9	I felt very confident using the system.	4.60 $\pm$ 0.52
Q10	I needed to learn a lot of things before I could get going with this system.	1.00 $\pm$ 0.00
Average percentile ranking		93.50 $\pm$ 6.03

mobile base, are dynamically adjusted to promote a more comfortable physical interaction with the user.

In order to systematically assess the effectiveness of the proposed approach, first, we conducted experiments involving 12 blindfolded human participants under two conditions: with and without the activation of the *Horizontal Pulling* unit. These trials were performed in a controlled laboratory environment free of obstacles that could affect the upper body. Results revealed that under baseline conditions, participants frequently deviated beyond the robot's safe boundary. In contrast, when lateral assistance was enabled, users consistently remained within the robot base's footprint (see Figure 7). We then evaluated the system's ability to provide effective guidance in the presence of dynamic obstacles through a second user study. The results demonstrated that the proposed approach was able to continuously replan the collision-free trajectory, enabling all participants to successfully reach the target without abrupt or erratic motions. Building upon this, the conducted proof-of-concept experiment in a narrow corridor has also validated the performance of the *Horizontal Pulling* strategy to prevent collisions while reaching a goal.

The second set of experiments focused on validating the full-body 3D safety of the user, specifically evaluating the effectiveness of the *Vertical Pulling* strategy in environments with elevated obstacles capable of impacting the upper body. Thanks to the proposed strategy that adjusts the end-effector height of the arm based on a RGB-D-based head tracking and 3D LiDAR-based obstacle detection, no collisions occurred in the user study conducted with 10 participants (see Figure 13). The state-based architecture of the *Vertical Pulling* unit enabled the system to adopt protective safety protocols when user correction failed and contact risk increased. In three cases, the system transitioned into the EMERGENCY state—halting base motion, increasing the arm vertical stiffness, and guiding the user more assertively back to a safe trajectory. Once safety was ensured, meaning the robot's assistance successfully repositioned the user away from potential danger, the robot

resumed its planned trajectory and reached the target. Notably, in the case of one participant, where there was no risk of collision because of the user's height, the proposed approach accurately recognized the situation and refrained from intervening unnecessarily, thereby prioritizing user comfort. The qualitative assessment conducted after the experiment, based on the System Usability Scale (SUS), yielded a score of 93.5%, surpassing the 85-point threshold for "excellent" usability, confirming strong user approval of the human-robot collaboration system.

Although the proposed approach demonstrates robust performance in guiding sighted participants who were blindfolded through complex environments, we acknowledge the importance of validating the system with visually impaired individuals. However, conducting such user studies was not feasible within the scope of this work due to the procedural requirements and extended timelines associated with obtaining ethical approval for research involving visually impaired participants. For this reason, we opted to conduct our experiments with sighted participants who were blindfolded to verify that the system performed consistently and reliably before involving visually impaired users.

Nevertheless, we expect the proposed 3D guidance approach to generalize effectively to visually impaired users, as it is composed of modular and configurable sub-units specifically designed to adapt based on user feedback. We also anticipate that evaluations with visually impaired participants may reveal mobility strategies or preferences that we may have previously overlooked, offering insights essential for improving the system. As part of our future work, we therefore plan to collaborate with visually impaired volunteers to assess system performance, gather qualitative feedback, and further refine the approach in alignment with the needs of the target population. In addition, several limitations of the current system will also be addressed. First, the integration of personalized assistive behaviors could improve user experience and comfort. In the current implementation, a fixed end-effector height was used to minimize inter-user variability during evaluation;

however, future studies could explore adaptive height regulation based on individual user profiles. Additionally, the adaptive impedance control strategy could be further refined by incorporating user-specific preferences, potentially learned through long-term interaction using data-driven approaches. Finally, to enable deployment in more dynamic and realistic settings, the navigation module could be extended to include semantic scene understanding (e.g., recognizing doors or signs) and socially-aware navigation, allowing the robot to interact appropriately in populated or shared environments.

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Supplemental material

Supplemental material for this article is available online.

Notes

1. Unit circle, set of angles $[-\pi, \pi)$.
2. The subject's data is documented as: mean $\pm$ standard deviation.

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